Learning to Lens

An Introduction to General Relativity and Strong Gravitational Lensing

with Mathematica

Based on the published works of

P. Schneider, J. Ehlers & E. E. Falco, *Gravitational Lenses* (1992)

R. Narayan & M. Bartelmann, *Lectures on Gravitational Lensing* (1997)

A. O. Petters, H. Levine & J. Wambsganss, *Singularity Theory and Gravitational Lensing* (2001)

S. Carroll, *Spacetime and Geometry* (2004)

A. B. Congdon & C. R. Keeton, *Principles of Gravitational Lensing* (2018)

M. Meneghetti, *Introduction to Gravitational Lensing* (2021)

P. Saha, D. Sluse, J. Wagner & L. L. R. Williams, *Essentials of Strong Gravitational Lensing* (2024)

— Instructor Edition (with Solutions) —

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About This Document

This tutorial suite was produced by **Rodrigo Córdova Rosado** (Harvard–Smithsonian Center for Astrophysics) with the assistance of **Claude** (Anthropic), an AI language model, via **Claude Code** — Anthropic's command-line tool for collaborative software engineering and technical writing.

The physics content in these notes draws from the following published textbooks and review articles:

- S. Carroll, *Spacetime and Geometry: An Introduction to General Relativity* (Addison Wesley, 2004)
- A. B. Congdon & C. R. Keeton, *Principles of Gravitational Lensing* (Springer, 2018)
- R. Narayan & M. Bartelmann, *Lectures on Gravitational Lensing* (1997; arXiv:astro-ph/9606001)
- P. Saha, D. Sluse, J. Wagner & L. L. R. Williams, *Essentials of Strong Gravitational Lensing*, Space Science Reviews **220**, 12 (2024)
- M. Meneghetti, *Introduction to Gravitational Lensing: With Python Examples* (Springer, 2021)
- P. Schneider, J. Ehlers & E. E. Falco, *Gravitational Lenses* (Springer, 1992)
- A. O. Petters, H. Levine & J. Wambsganss, *Singularity Theory and Gravitational Lensing* (Birkhäuser, 2001)

The workflow was as follows: the author directed Claude to co-write the LaTeX lecture notes, Wolfram Mathematica scripts, and interactive notebooks, drawing on the reference texts above. Claude drafted the text and code; the author reviewed, revised, and approved all content.

Importantly, all mathematical derivations and symbolic results in these notes have been verified deterministically using Wolfram Mathematica's computer algebra system — not generated probabilistically by an LLM. Every equation, identity, and worked solution was checked by running the companion `.wl` scripts (via `wolframscript`), which perform exact symbolic computation. The Mathematica verification outputs are reproducible and accompany each module in the repository.

The AI assistant was used for:

- Drafting expository prose and pedagogical explanations
- Structuring the curriculum and organizing modules
- Writing Mathematica code (subsequently executed and verified)
- Formatting LaTeX and managing cross-references

The AI assistant was **not** used as a source of mathematical truth. All physics content derives from the published works listed above, and all symbolic computations are verified by Mathematica.

This is a **living document**. Additional details, examples, and lessons will be added over time as the author continues to learn and teach this material. The most current version is always available at: https://github.com/rodelcr/Learning_to_Lens

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Notation Guide

Symbol	Meaning	Units / Notes
Coordinates and Angles		
$\boldsymbol{\theta}$	Image position (angular)	radians or arcsec
$\boldsymbol{\beta}$	Source position (angular)	radians or arcsec
$\boldsymbol{\alpha}$	Reduced deflection angle	radians
$\hat{\alpha}$	Physical deflection angle	radians
$\boldsymbol{\xi}$	Position in lens plane	physical (kpc, Mpc)
Distances		
D_{d}	Observer–lens angular diameter distance	Mpc
D_{s}	Observer–source angular diameter distance	Mpc
D_{ds}	Lens–source angular diameter distance	Mpc
Lensing Quantities		
θ_{E}	Einstein radius	arcsec
Σ_{cr}	Critical surface mass density	$\mathrm{g\,cm^{-2}}$ or $M_\odot\,\mathrm{pc}^{-2}$
κ	Convergence $= \Sigma/\Sigma_{\mathrm{cr}}$	dimensionless
γ	Shear magnitude	dimensionless
ψ	Effective lensing potential	dimensionless
μ	Magnification	dimensionless
$\mathcal{A}$	Jacobian (inverse magnification) matrix	2×2
Cosmology		
H_0	Hubble constant	$\mathrm{km\,s^{-1}\,Mpc^{-1}}$
Ω_{m}	Matter density parameter	dimensionless
Ω_Λ	Dark energy density parameter	dimensionless
GR / Metric		
$g_{\mu\nu}$	Metric tensor	
$\Gamma^{\alpha}{}_{\mu\nu}$	Christoffel symbols	
$R^{\alpha}{}_{\beta\mu\nu}$	Riemann curvature tensor	
$R_{\mu\nu}$	Ricci tensor	
R	Ricci scalar	

Mathematica files are cross-linked throughout these notes. Links formatted as `filename.nb` or `filename.wl` point to companion Mathematica notebooks and scripts in the `Mathematica/`

directory of this repository.

Part I

General Relativity Foundations

Chapter 1

Special Relativity and Tensor Basics

> *General relativity (GR) is Einstein's theory of space, time, and gravitation. At heart it is a very simple subject . . . The essential idea is perfectly straightforward: while most forces of nature are represented by fields defined on spacetime, gravity is inherent in spacetime itself.* — Sean Carroll, *Spacetime and Geometry*

Before we can understand how gravity bends light, we need the language of special relativity and tensors. This chapter reviews the foundations: Minkowski spacetime, Lorentz transformations, four-vectors, and the index notation that will carry us through the rest of these notes. The emphasis is on building fluency with the notation and the key physical ideas — time dilation, length contraction, the invariant interval — so that the transition to curved spacetime in later modules feels natural.

Companion Mathematica files:

- `lorentz_transforms.wl` — symbolic construction and verification of Lorentz boost matrices
- `four_vectors.nb` — interactive visualizations of worldlines, light cones, and time dilation

1.1 From Newton to Einstein

In Newtonian mechanics, space and time are independent. The gravitational force between two masses M and m separated by a distance r is (cf. Carroll eq. 1.1)

$$\mathbf{F} = -\frac{GMm}{r^2}\,\hat{\mathbf{r}}\,, \tag{1.1}$$

and the gravitational potential Φ satisfies Poisson's equation,

$$\nabla^2\Phi = 4\pi G\rho\,. \tag{1.2}$$

In GR, both of these are replaced by statements about the curvature of spacetime. The key equation we will eventually arrive at is Einstein's field equation (Carroll eq. 1.5):

$$R_{\mu\nu} - \tfrac{1}{2}R\,g_{\mu\nu} = 8\pi G\,T_{\mu\nu}\,. \tag{1.3}$$

But we are getting ahead of ourselves. For now, we focus on *flat* spacetime — the arena of special relativity — and learn the tensor language needed to write down equations like (1.3).

1.2 Spacetime and the Invariant Interval

Special relativity (SR) unifies space and time into a single four-dimensional continuum called **Minkowski space**. An individual point in spacetime is an **event**, and the path of a particle through spacetime is its **worldline**.

Unlike Newtonian mechanics, SR has no absolute notion of simultaneity. Instead, the fundamental invariant is the **spacetime interval** between two events (Carroll eq. 1.10):

$$\boxed{(\Delta s)^2 = -(\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2} \tag{1.4}$$

where we work in **natural units** with $c = 1$ throughout this chapter (we will restore factors of c when needed for astrophysical applications).

Remark 1.2.1. We adopt the **mostly-plus** sign convention $(-, +, +, +)$, following Carroll. Some references (especially in particle physics) use the opposite convention $(+, -, -, -)$. Be careful when comparing equations across textbooks.

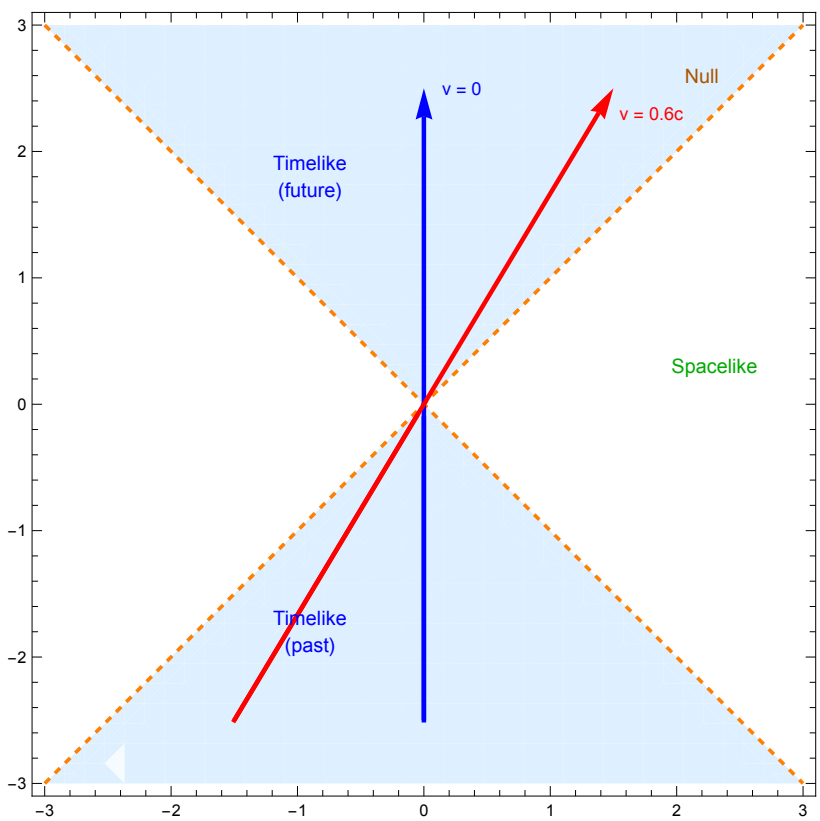


Figure 1.1: Spacetime diagram in 1+1 dimensions showing the light cone structure. The shaded (blue) region is timelike; the dashed orange lines are the light cone (null); the unshaded region is spacelike. The blue arrow is the worldline of a stationary particle; the red arrow is a particle moving at $v = 0.6c$. (cf. Carroll Fig. 1.5)

The interval classifies the separation between events:

- $(\Delta s)^2 < 0$: **timelike** separation (events can be causally connected)

- $(\Delta s)^2 = 0$: **null** or **lightlike** separation (connected by a light ray)
- $(\Delta s)^2 > 0$: **spacelike** separation (no causal connection possible)

The **proper time** $\Delta\tau$ along a timelike path is defined as (Carroll eq. 1.17)

$$(\Delta\tau)^2 = -(\Delta s)^2 = (\Delta t)^2 - (\Delta x)^2 - (\Delta y)^2 - (\Delta z)^2 \,. \tag{1.5}$$

This is the time measured by a clock traveling along the worldline. For an infinitesimal displacement, we write the **line element** (Carroll eq. 1.20):

$$ds^2 = \eta_{\mu\nu}\, dx^\mu\, dx^\nu \,, \tag{1.6}$$

where we have introduced the **Minkowski metric** and the **Einstein summation convention** (repeated indices are summed over).

1.3 Index Notation and the Minkowski Metric

Spacetime coordinates are labeled by Greek indices running from 0 to 3 (Carroll eq. 1.12):

$$x^\mu : \quad x^0 = t, \quad x^1 = x, \quad x^2 = y, \quad x^3 = z \,. \tag{1.7}$$

Latin indices $i, j, k, \ldots$ run over the spatial components $1, 2, 3$ only.

The **Minkowski metric** is the 4×4 matrix (Carroll eq. 1.15):

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \,. \tag{1.8}$$

Its inverse, $\eta^{\mu\nu}$, has the same components. The metric is used to **raise and lower indices**:

$$V_\mu = \eta_{\mu\nu}\, V^\nu, \qquad \omega^\mu = \eta^{\mu\nu}\, \omega_\nu \,. \tag{1.9}$$

Note that in Minkowski space this simply flips the sign of the time component: $V_0 = -V^0$, while $V_i = V^i$.

1.4 Lorentz Transformations

The coordinate transformations that leave the interval (1.6) invariant are called **Lorentz transformations**. They act on the coordinates as (Carroll eq. 1.25)

$$x^{\mu'} = \Lambda^{\mu'}{}_\nu\, x^\nu \,, \tag{1.10}$$

where the matrix Λ satisfies the defining condition (Carroll eq. 1.28):

$$\boxed{\eta = \Lambda^{\mathrm{T}}\, \eta\, \Lambda} \tag{1.11}$$

or equivalently $\eta_{\rho\sigma} = \Lambda^{\mu'}{}_\rho\, \Lambda^{\nu'}{}_\sigma\, \eta_{\mu'\nu'}$.

1.4.1 Rotations

A spatial rotation in the x–y plane by angle θ is (Carroll eq. 1.31):

$$\Lambda^{\mu'}{}_{\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & \sin\theta & 0 \\ 0 & -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} . \tag{1.12}$$

1.4.2 Boosts

A **Lorentz boost** along the x-axis with rapidity parameter ϕ is (Carroll eq. 1.32):

$$\Lambda^{\mu'}{}_{\nu} = \begin{pmatrix} \cosh\phi & -\sinh\phi & 0 & 0 \\ -\sinh\phi & \cosh\phi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} . \tag{1.13}$$

Using $v = \tanh\phi$ and $\gamma = (1 - v^2)^{-1/2} = \cosh\phi$, we recover the familiar form (Carroll eq. 1.35):

$$\begin{aligned} t' &= \gamma(t - vx) \,, \\ x' &= \gamma(x - vt) \,. \end{aligned} \tag{1.14}$$

Exercise 1.4.1. Verify that the boost matrix (1.13) satisfies the Lorentz condition (1.11) by explicit matrix multiplication. Then confirm this symbolically using `lorentz_transforms.wl`.

1.4.3 Physical Consequences

The boost equations (1.14) immediately lead to the classic relativistic effects:

- **Time dilation:** A clock at rest in S' (i.e., $\Delta x' = 0$) ticks at intervals $\Delta t = \gamma\, \Delta t'$.
- **Length contraction:** A rod at rest in S' with proper length L_0 has measured length $L = L_0/\gamma$ in S.
- **Relativity of simultaneity:** Events simultaneous in S' ($\Delta t' = 0$) satisfy $\Delta t = \gamma v \Delta x$ in S.

Exercise 1.4.2. (Twin paradox.) An observer travels from event $A = (0, 0)$ to $B' = (\Delta t/2,\, v\Delta t/2)$ and back to $C = (\Delta t, 0)$ at constant speed v, while a stationary twin travels from A directly to C. Show that the traveling twin ages by $\Delta\tau_{AB'C} = \sqrt{1 - v^2}\, \Delta t$ while the stationary twin ages by $\Delta\tau_{ABC} = \Delta t$. (cf. Carroll eqs. 1.18–1.19. Visualize with `four_vectors.nb`.)

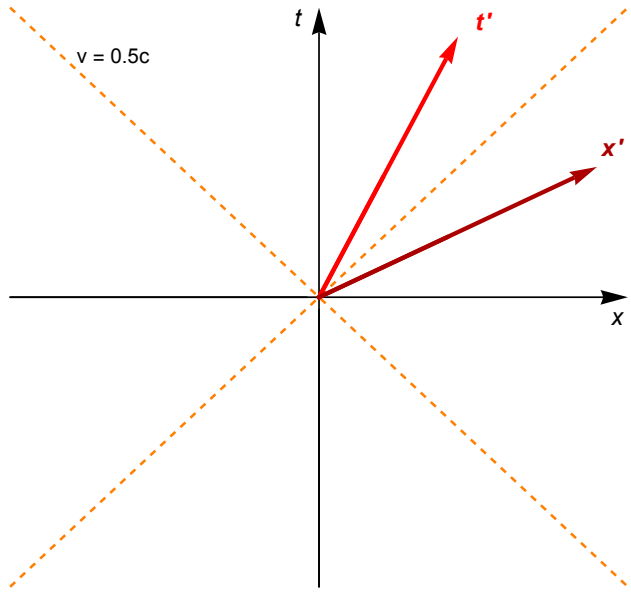


Figure 1.2: A Lorentz boost with $v = 0.5c$ tilts both the time axis (t', red) and the space axis (x', dark red) toward the light cone. The light cone itself (orange dashed) is invariant under boosts. (cf. Carroll Fig. 1.7)

1.5 Four-Vectors

A **four-vector** V^μ is a set of four components that transforms under Lorentz transformations as (Carroll eq. 1.39):

$$V^{\mu'} = \Lambda^{\mu'}{}_{\nu} V^\nu \,. \tag{1.15}$$

Key examples:

Four-position: $x^\mu = (t, x, y, z)$.

Four-velocity: For a massive particle with worldline $x^\mu(\tau)$, the four-velocity is

$$U^\mu = \frac{dx^\mu}{d\tau} = \gamma(1, v^x, v^y, v^z) \,. \tag{1.16}$$

Its norm is always $\eta_{\mu\nu} U^\mu U^\nu = -1$ (in units with $c = 1$).

Four-momentum: $p^\mu = m\, U^\mu = (E, p^x, p^y, p^z)$, where $E = \gamma m$ is the relativistic energy and $\mathbf{p} = \gamma m\, \mathbf{v}$ is the three-momentum. The invariant norm gives the **energy–momentum relation**:

$$\boxed{\eta_{\mu\nu}\, p^\mu p^\nu = -E^2 + |\mathbf{p}|^2 = -m^2} \tag{1.17}$$

or, restoring c: $E^2 = (pc)^2 + (mc^2)^2$.

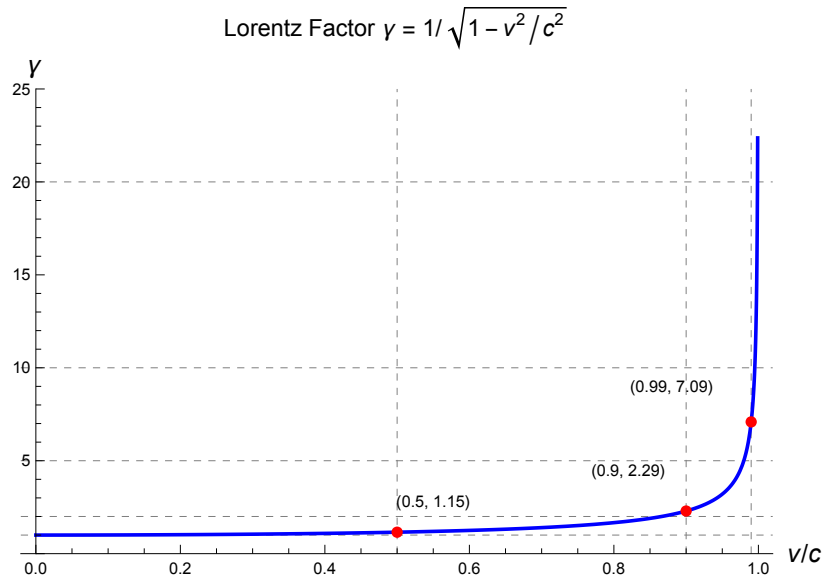


Figure 1.3: The Lorentz factor $\gamma = (1 - v^2/c^2)^{-1/2}$ as a function of velocity. Time dilation and length contraction both scale with γ. Note the dramatic increase as $v \to c$.

For a **photon** ($m = 0$), $E = |\mathbf{p}|$ and the four-momentum is null: $p^\mu p_\mu = 0$. This will be crucial when we trace light rays through curved spacetime in later modules.

1.6 Dual Vectors (One-Forms)

Associated with every vector space is a **dual vector space** (or cotangent space T_p^*). A dual vector (or **one-form**) ω_μ transforms under the *inverse* Lorentz transformation (Carroll eq. 1.50):

$$\omega_{\mu'} = \Lambda^\nu{}_{\mu'}\, \omega_\nu \,. \tag{1.18}$$

The prototypical one-form is the **gradient** of a scalar field ϕ (Carroll eq. 1.54):

$$(\mathrm{d}\phi)_\mu = \partial_\mu \phi \equiv \frac{\partial \phi}{\partial x^\mu} \,. \tag{1.19}$$

The contraction of a vector with a dual vector is a Lorentz scalar: $\omega_\mu V^\mu \in \mathbf{R}$.

1.7 Tensors

A **tensor** of type (k, l) is a multilinear map that takes k one-forms and l vectors and produces a real number (Carroll eq. 1.56). In components, the transformation law is (Carroll eq. 1.63):

$$T^{\mu'_1 \cdots \mu'_k}{}_{\nu'_1 \cdots \nu'_l} = \Lambda^{\mu'_1}{}_{\alpha_1} \cdots \Lambda^{\mu'_k}{}_{\alpha_k} \Lambda^{\beta_1}{}_{\nu'_1} \cdots \Lambda^{\beta_l}{}_{\nu'_l} T^{\alpha_1 \cdots \alpha_k}{}_{\beta_1 \cdots \beta_l} \,. \tag{1.20}$$

1.7.1 Important Operations

- **Contraction:** Summing over one upper and one lower index reduces the rank by $(1, 1)$ (Carroll eq. 1.70): $S^{\mu\rho}{}_{\sigma} = T^{\mu\nu\rho}{}_{\sigma\nu}$.
- **Raising/lowering:** Use the metric to move indices (Carroll eq. 1.72): $T^{\alpha\beta\mu}{}_{\delta} = \eta^{\mu\gamma}\, T^{\alpha\beta}{}_{\gamma\delta}$.
- **Symmetrization/antisymmetrization:** Round brackets denote symmetrization; square brackets denote antisymmetrization (Carroll eqs. 1.79–1.80).

1.7.2 Key Tensors in Minkowski Space

- **Metric:** $\eta_{\mu\nu}$ — a symmetric $(0, 2)$ tensor.
- **Kronecker delta:** $\delta^{\mu}{}_{\nu}$ — the identity map.
- **Levi-Civita symbol:** $\tilde{\epsilon}_{\mu\nu\rho\sigma}$ (Carroll eq. 1.68) — totally antisymmetric, with $\tilde{\epsilon}_{0123} = +1$.
- **Electromagnetic field strength:** $F_{\mu\nu} = -F_{\nu\mu}$ (Carroll eq. 1.69) — a $(0, 2)$ antisymmetric tensor encoding $\mathbf{E}$ and $\mathbf{B}$.

1.8 The Metric as the Central Object

The key insight that connects SR to GR is that the metric is the fundamental object describing the geometry of spacetime. In SR, the metric is the constant $\eta_{\mu\nu}$. In GR, the metric becomes a field $g_{\mu\nu}(x)$ that varies from point to point, encoding the curvature of spacetime due to matter and energy.

Everything we need for gravitational lensing — the bending of light, the distortion of images, the time delays between multiple images — follows from understanding how the metric determines the paths of light rays. That story begins in the next module, where we introduce curved spacetime and the geodesic equation.

1.9 Energy and Momentum in SR

The **stress-energy tensor** $T^{\mu\nu}$ encodes the energy and momentum content of matter and fields. For a perfect fluid with energy density ρ, pressure p, and four-velocity U^{μ}:

$$T^{\mu\nu} = (\rho + p)\, U^{\mu} U^{\nu} + p\, \eta^{\mu\nu}\,. \tag{1.21}$$

Conservation of energy and momentum is expressed as

$$\partial_{\mu} T^{\mu\nu} = 0\,. \tag{1.22}$$

This will generalize to $\nabla_{\mu} T^{\mu\nu} = 0$ in curved spacetime, and the stress-energy tensor will appear as the source term on the right-hand side of Einstein's field equation (1.3).

1.10 Summary

- Spacetime is a four-dimensional continuum; the invariant quantity is the interval $ds^2 = \eta_{\mu\nu}\, dx^\mu dx^\nu$.
- Lorentz transformations ($x^{\mu'} = \Lambda^{\mu'}{}_\nu x^\nu$) leave the interval invariant. Boosts mix space and time.
- Four-vectors and tensors are classified by how they transform under Lorentz transformations.
- The metric $\eta_{\mu\nu}$ is the central object — it defines distances, raises/lowers indices, and generalizes to $g_{\mu\nu}$ in GR.
- Photons travel on null paths ($ds^2 = 0$), which will be the starting point for understanding gravitational lensing.

1.11 Problems

Exercise 1.11.1. Classify the following pairs of events as timelike, spacelike, or null separated:

(a) $A = (0, 0, 0, 0)$, $B = (2, 1, 1, 0)$

(b) $A = (0, 0, 0, 0)$, $B = (1, 1, 0, 0)$

(c) $A = (0, 0, 0, 0)$, $B = (3, 1, 2, 2)$

Check your answers with Mathematica.

Exercise 1.11.2. Show that two successive boosts along the x-axis with rapidities ϕ_1 and ϕ_2 are equivalent to a single boost with rapidity $\phi = \phi_1 + \phi_2$. Relate this to the velocity addition formula $v = (v_1 + v_2)/(1 + v_1 v_2)$.

Exercise 1.11.3. Prove that the four-velocity of any massive particle satisfies $U^\mu U_\mu = -1$ (in units with $c = 1$).

Exercise 1.11.4. A photon with energy E in frame S is observed in frame S' moving with velocity v along the x-axis.

(a) Show that the photon energy in S' is $E' = \gamma E(1 - v\cos\theta)$, where θ is the angle between the photon's direction and the x-axis in frame S.

(b) Specialize to $\theta = 0$ (head-on) and $\theta = \pi/2$ (transverse) and interpret physically.

Exercise 1.11.5. The electromagnetic field strength tensor has components (Carroll eq. 1.69):

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_1 & -E_2 & -E_3 \\ E_1 & 0 & B_3 & -B_2 \\ E_2 & -B_3 & 0 & B_1 \\ E_3 & B_2 & -B_1 & 0 \end{pmatrix}$$

(a) Show that $F_{\mu\nu}$ is antisymmetric.

(b) Compute $F^{\mu\nu} = \eta^{\mu\alpha}\eta^{\nu\beta}F_{\alpha\beta}$ and verify that the signs of the electric field components change.

(c) Compute the Lorentz scalar $F_{\mu\nu}F^{\mu\nu}$ and express it in terms of $\mathbf{E}$ and $\mathbf{B}$.

1.12 Solutions to Exercises

Full worked solutions are also available in the Mathematica script `problems_01a.wl`, which verifies each answer symbolically.

Exercise 1.11.1: Interval Classification

We compute $(\Delta s)^2 = -(\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2$:

(a) $(\Delta s)^2 = -(2)^2 + (1)^2 + (1)^2 + (0)^2 = -4 + 1 + 1 = -2 < 0 \Rightarrow$ **Timelike**. A massive particle can travel between these events.

(b) $(\Delta s)^2 = -(1)^2 + (1)^2 + (0)^2 + (0)^2 = 0 \Rightarrow$ **Null**. A light ray connects these events.

(c) $(\Delta s)^2 = -(3)^2 + (1)^2 + (2)^2 + (2)^2 = -9 + 1 + 4 + 4 = 0 \Rightarrow$ **Null**. Again lightlike: $|\Delta \mathbf{x}|^2 = 9 = (\Delta t)^2$.

Exercise 1.11.2: Boost Composition

The boost matrix for rapidity ϕ is $\Lambda(\phi) = \left(\begin{smallmatrix} \cosh\phi & -\sinh\phi \\ -\sinh\phi & \cosh\phi \end{smallmatrix}\right)$ (showing only the t–x block). Computing the product:

$$\begin{aligned}
\Lambda(\phi_1)\,\Lambda(\phi_2) &= \begin{pmatrix} \cosh\phi_1\cosh\phi_2 + \sinh\phi_1\sinh\phi_2 & -\cosh\phi_1\sinh\phi_2 - \sinh\phi_1\cosh\phi_2 \\ -\sinh\phi_1\cosh\phi_2 - \cosh\phi_1\sinh\phi_2 & \sinh\phi_1\sinh\phi_2 + \cosh\phi_1\cosh\phi_2 \end{pmatrix} \\
&= \begin{pmatrix} \cosh(\phi_1+\phi_2) & -\sinh(\phi_1+\phi_2) \\ -\sinh(\phi_1+\phi_2) & \cosh(\phi_1+\phi_2) \end{pmatrix} = \Lambda(\phi_1+\phi_2)\,,
\end{aligned}$$

using the hyperbolic addition formulas.

For the velocity addition formula, we use $v = \tanh\phi$:

$$v = \tanh(\phi_1+\phi_2) = \frac{\tanh\phi_1 + \tanh\phi_2}{1 + \tanh\phi_1\tanh\phi_2} = \frac{v_1 + v_2}{1 + v_1 v_2}\,.$$

Note: for $v_1 = v_2 = 1$ (speed of light), $v = 2/2 = 1$, confirming that the speed of light is invariant.

Exercise 1.11.3: Four-Velocity Norm

The four-velocity is $U^\mu = \gamma(1, v^1, v^2, v^3)$ where $\gamma = (1-v^2)^{-1/2}$ and $v^2 = (v^1)^2 + (v^2)^2 + (v^3)^2$. Then:

$$\begin{aligned}
U^\mu U_\mu = \eta_{\mu\nu} U^\mu U^\nu &= \gamma^2\left[-(1)^2 + (v^1)^2 + (v^2)^2 + (v^3)^2\right] \\
&= \gamma^2(-1 + v^2) = \frac{-(1-v^2)}{1-v^2} = \boxed{-1}\,.
\end{aligned}$$

Exercise 1.11.4: Photon Energy (Relativistic Doppler)

(a) A photon with energy E propagating at angle θ to the x-axis has four-momentum $p^\mu = E(1, \cos\theta, \sin\theta, 0)$. Applying the boost Λ along x:

$$E' = p^{0'} = \gamma(p^0 - v\,p^1) = \gamma E(1 - v\cos\theta)\,.$$

(b) Special cases:

- $\theta = 0$ (photon along $+x$, same direction as boost): $E' = \gamma E(1-v) = E\sqrt{\frac{1-v}{1+v}}$. This is a **redshift** ($E' < E$). The observer recedes from the source.
- $\theta = \pi/2$ (transverse): $E' = \gamma E = E/\sqrt{1-v^2}$. This is the **transverse Doppler effect** — a purely relativistic blueshift with no classical analogue, arising from time dilation.

Exercise 1.11.5: Electromagnetic Field Strength Tensor

(a) By inspection, $F_{\mu\nu} = -F_{\nu\mu}$ (the matrix is antisymmetric: the diagonal is zero, and off-diagonal elements differ by a sign).

(b) Raising both indices with the metric: $F^{\mu\nu} = \eta^{\mu\alpha}\eta^{\nu\beta}F_{\alpha\beta}$. In matrix form: $F^{\mu\nu} = \eta \cdot F_{\mu\nu} \cdot \eta$. The result is:

$$F^{\mu\nu} = \begin{pmatrix} 0 & E_1 & E_2 & E_3 \\ -E_1 & 0 & B_3 & -B_2 \\ -E_2 & -B_3 & 0 & B_1 \\ -E_3 & B_2 & -B_1 & 0 \end{pmatrix}$$

The electric field components flip sign (from raising the time index), while the magnetic components are unchanged.

(c) The Lorentz scalar:

$$\begin{aligned} F_{\mu\nu}F^{\mu\nu} &= 2\big(-E_1^2 - E_2^2 - E_3^2 + B_1^2 + B_2^2 + B_3^2\big) \\ &= 2(|\mathbf{B}|^2 - |\mathbf{E}|^2)\,. \end{aligned}$$

This is a Lorentz invariant: all inertial observers agree on the value of $|\mathbf{B}|^2 - |\mathbf{E}|^2$.

Chapter 2

Differential Geometry and the Metric Tensor

> *The appropriate mathematical structure used to describe curvature is that of a differentiable manifold: essentially, a kind of set that looks locally like flat space, but might have a very different global geometry.* — Sean Carroll, *Spacetime and Geometry*

In Module 1a we worked exclusively in flat Minkowski spacetime. To describe gravity, we must allow spacetime to be *curved*. This chapter introduces the mathematical machinery of differential geometry — manifolds, the general metric tensor, the covariant derivative, Christoffel symbols, and the curvature tensors — that forms the foundation of general relativity and, ultimately, gravitational lensing.

Companion Mathematica files:

- `christoffel_symbols.wl` — general-purpose symbolic computation of Christoffel symbols and curvature tensors from any input metric
- `curved_surfaces.nb` — interactive visualizations of parallel transport, geodesics, and curvature on 2D surfaces

2.1 Manifolds and Coordinate Charts

A **manifold** is a space that may be curved globally but looks like $\mathbb{R}^n$ in every sufficiently small neighborhood. More precisely, an n-dimensional manifold M is a topological space equipped with a collection of **coordinate charts** — invertible maps $\phi : U \to \mathbb{R}^n$ from open subsets $U \subset M$ — that cover all of M and are smoothly compatible on overlaps (Carroll Sec. 2.2).

Examples of manifolds include:

- $\mathbb{R}^n$ (trivially)
- The 2-sphere S^2 (the surface of a ball)
- The 2-torus T^2 (the surface of a doughnut)
- Spacetime itself (a 4-dimensional Lorentzian manifold)

A key fact: in general, *no single coordinate system can cover an entire manifold.* The sphere S^2, for instance, requires at least two charts (e.g., stereographic projections from the north and south poles; Carroll eqs. 2.9–2.10). We will always work locally in a single chart, keeping in mind that our results are coordinate-independent.

2.2 The General Metric Tensor

On a manifold, we need a way to measure distances and angles. This is provided by the **metric tensor** $g_{\mu\nu}(x)$, a symmetric, nondegenerate $(0, 2)$ tensor field. The line element is (cf. Carroll eq. 1.20, now generalized):

$$\boxed{ds^2 = g_{\mu\nu}(x)\, dx^\mu\, dx^\nu} \tag{2.1}$$

In flat Minkowski space, $g_{\mu\nu} = \eta_{\mu\nu}$ everywhere. In curved spacetime, $g_{\mu\nu}$ varies from point to point and encodes the gravitational field.

The **inverse metric** $g^{\mu\nu}$ is defined by

$$g^{\mu\lambda} g_{\lambda\nu} = \delta^\mu{}_\nu\,. \tag{2.2}$$

As before, the metric raises and lowers indices: $V_\mu = g_{\mu\nu} V^\nu$, $V^\mu = g^{\mu\nu} V_\nu$.

Example 2.2.1. The metric on the 2-sphere of radius R in spherical coordinates (θ, ϕ) is

$$ds^2 = R^2\big(d\theta^2 + \sin^2\theta\, d\phi^2\big)\,, \tag{2.3}$$

so $g_{\theta\theta} = R^2$, $g_{\phi\phi} = R^2 \sin^2\theta$, and all other components vanish.

Example 2.2.2. The flat plane in polar coordinates (r, ϕ) has metric

$$ds^2 = dr^2 + r^2\, d\phi^2\,. \tag{2.4}$$

Despite the coordinate-dependent appearance, this space is flat — the curvature tensors vanish. We will verify this in Mathematica.

2.3 Vectors and Tensors on Manifolds

On a general manifold, vectors live in the **tangent space** T_p at each point p. A vector V at p is identified with a directional derivative operator along a curve $x^\mu(\lambda)$ through p (Carroll Sec. 2.3):

$$V = V^\mu \partial_\mu\,, \tag{2.5}$$

where $\partial_\mu \equiv \partial/\partial x^\mu$ form a **coordinate basis** for T_p.

Under a coordinate transformation $x^\mu \to x^{\mu'}$, vector components transform as (Carroll eq. 2.19):

$$V^{\mu'} = \frac{\partial x^{\mu'}}{\partial x^\nu}\, V^\nu\,. \tag{2.6}$$

This generalizes the Lorentz transformation law from Module 1a to arbitrary coordinate changes.

Dual vectors (one-forms) and general tensors transform analogously, with one factor of $\partial x^{\mu'}/\partial x^\nu$ for each upper index and $\partial x^\nu/\partial x^{\mu'}$ for each lower index.

2.4 The Covariant Derivative

Ordinary partial derivatives of tensors are *not* tensors on a curved manifold: the transformation of $\partial_\mu V^\nu$ picks up an extra, non-tensorial term from the coordinate dependence of the basis vectors (Carroll Sec. 3.2).

To fix this, we introduce the **covariant derivative** ∇_μ, which adds a correction term involving the **connection coefficients** $\Gamma^\rho{}_{\mu\sigma}$ (Carroll eq. 3.5):

$$\boxed{\nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma^\nu{}_{\mu\lambda} V^\lambda} \tag{2.7}$$

For a one-form (Carroll eq. 3.16):

$$\nabla_\mu \omega_\nu = \partial_\mu \omega_\nu - \Gamma^\lambda{}_{\mu\nu} \omega_\lambda \,. \tag{2.8}$$

The general rule for a tensor of arbitrary rank: $+\Gamma$ for each upper index, $-\Gamma$ for each lower index (Carroll eq. 3.17).

The connection coefficients are *not* tensors — they transform inhomogeneously (Carroll eq. 3.10) — but the covariant derivative $\nabla_\mu V^\nu$ *is* a tensor.

2.5 Christoffel Symbols

In GR we demand two additional properties of the connection:

1. **Torsion-free:** $\Gamma^\lambda{}_{\mu\nu} = \Gamma^\lambda{}_{\nu\mu}$
2. **Metric compatibility:** $\nabla_\rho g_{\mu\nu} = 0$

These uniquely determine the **Christoffel symbols** (also called the Levi-Civita connection; Carroll eq. 3.27):

$$\boxed{\Gamma^\sigma{}_{\mu\nu} = \frac{1}{2} g^{\sigma\rho}\big(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}\big)} \tag{2.9}$$

This is one of the most important formulas in GR. Given any metric, we can compute the Christoffel symbols and from them all of the geometry.

Example 2.5.1. For the plane in polar coordinates, $ds^2 = dr^2 + r^2 d\phi^2$ (Example 2.2.2), the only nonvanishing Christoffel symbols are (Carroll eqs. 3.29–3.31):

$$\Gamma^r{}_{\phi\phi} = -r \,, \qquad \Gamma^\phi{}_{r\phi} = \Gamma^\phi{}_{\phi r} = \frac{1}{r} \,. \tag{2.10}$$

We derive these and more general cases in `christoffel_symbols.wl`.

2.6 Parallel Transport

In flat space, we can freely slide a vector from one point to another while keeping it "constant." On a curved manifold, this is no longer unambiguous — the result depends on the *path taken*.

The notion of keeping a vector constant along a curve $x^\mu(\lambda)$ is formalized as **parallel transport**: a vector V^μ is parallel-transported if (Carroll eq. 3.40)

$$\frac{dV^\mu}{d\lambda} + \Gamma^\mu{}_{\rho\sigma} \frac{dx^\rho}{d\lambda} V^\sigma = 0\,. \tag{2.11}$$

An important consequence of metric compatibility is that parallel transport preserves the inner product: if V^μ and W^μ are both parallel-transported along a curve, then $g_{\mu\nu}V^\mu W^\nu = \text{const}$ (Carroll eq. 3.42).

On the 2-sphere, parallel transporting a vector around a closed loop results in a rotation of the vector — the angle of rotation is proportional to the enclosed area and the curvature (Figure 2.1).

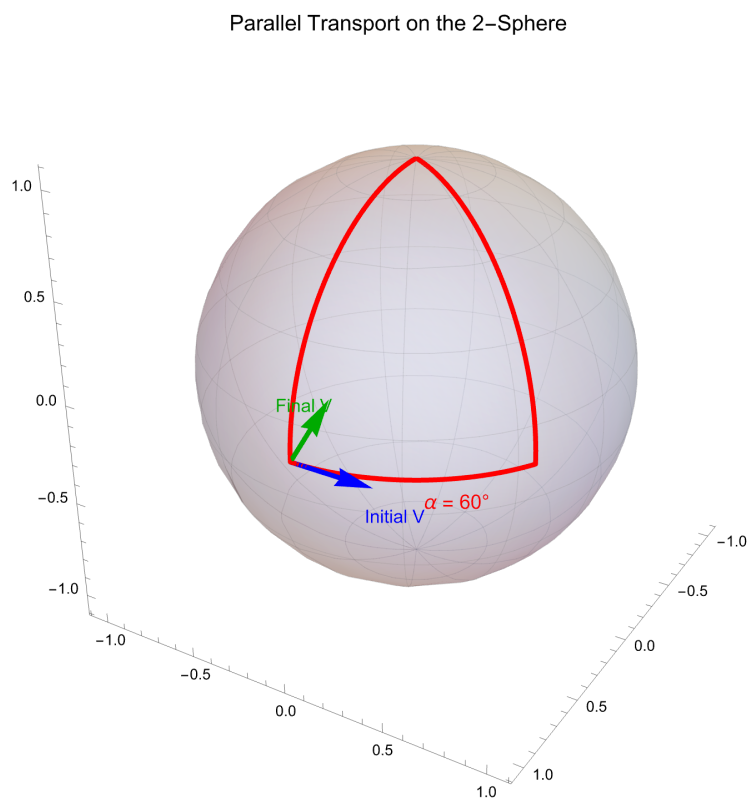


Figure 2.1: Parallel transport on the 2-sphere. A vector (blue) initially pointing east at the equator is transported north to the pole along $\phi = 0$, then south along $\phi = 60$ back to the equator. The final vector (green) has rotated by $\alpha = 60$ — equal to the solid angle enclosed by the path. This rotation is a direct manifestation of curvature. (cf. Carroll Fig. 3.2)

This is also visualized interactively in `curved_surfaces.nb`.

2.7 The Geodesic Equation

A **geodesic** is the curved-space generalization of a straight line: it is the path that parallel-transports its own tangent vector. Setting $V^\mu = dx^\mu/d\lambda$ in (2.11), we obtain the **geodesic**

equation (Carroll eq. 3.44):

$$\boxed{\frac{d^2x^\mu}{d\lambda^2} + \Gamma^\mu{}_{\rho\sigma}\frac{dx^\rho}{d\lambda}\frac{dx^\sigma}{d\lambda} = 0} \tag{2.12}$$

This can equivalently be derived by extremizing the proper time/interval functional (Carroll Sec. 3.3, eqs. 3.45–3.55; Congdon & Keeton eq. 3.60).

Key properties:

- **Timelike geodesics** ($ds^2 < 0$): paths of freely falling massive particles; parameterized by proper time τ.
- **Null geodesics** ($ds^2 = 0$): paths of light rays; parameterized by an affine parameter λ.
- **Spacelike geodesics** ($ds^2 > 0$): shortest spatial distances.
- The character (timelike/null/spacelike) is preserved along the geodesic (Carroll Sec. 3.4).

In flat space with Cartesian coordinates, $\Gamma^\mu{}_{\rho\sigma} = 0$, so the geodesic equation reduces to $d^2x^\mu/d\lambda^2 = 0$ — a straight line, as expected.

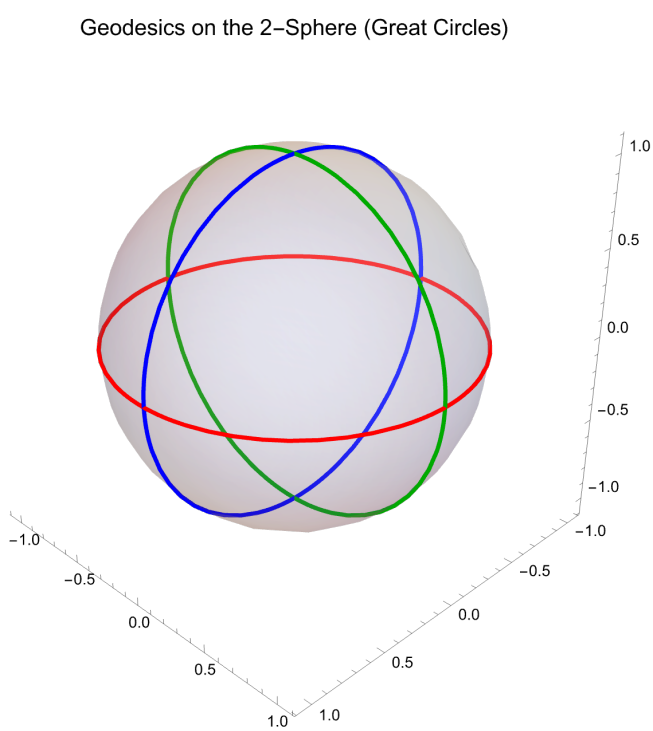


Figure 2.2: Geodesics on the 2-sphere are great circles. Three orthogonal great circles are shown, illustrating that the shortest path between any two points on a sphere follows a great circle arc.

2.8 Curvature Tensors

The **Riemann curvature tensor** $R^{\rho}{}_{\sigma\mu\nu}$ measures how much parallel transport around a closed loop rotates a vector. It is defined by (Carroll eq. 3.4):

$$\boxed{R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu \Gamma^{\rho}{}_{\nu\sigma} - \partial_\nu \Gamma^{\rho}{}_{\mu\sigma} + \Gamma^{\rho}{}_{\mu\lambda}\,\Gamma^{\lambda}{}_{\nu\sigma} - \Gamma^{\rho}{}_{\nu\lambda}\,\Gamma^{\lambda}{}_{\mu\sigma}} \tag{2.13}$$

A manifold is flat if and only if $R^{\rho}{}_{\sigma\mu\nu} = 0$ everywhere. This is the mathematical statement that curvature is characterized entirely by the Riemann tensor.

From the Riemann tensor we construct:

- **Ricci tensor** (contraction on 1st and 3rd indices):

$$R_{\mu\nu} = R^{\lambda}{}_{\mu\lambda\nu} \tag{2.14}$$

- **Ricci scalar** (trace of the Ricci tensor):

$$R = g^{\mu\nu} R_{\mu\nu} \tag{2.15}$$

These are the quantities that appear in Einstein's field equation

$$R_{\mu\nu} - \frac{1}{2} R\, g_{\mu\nu} = 8\pi G\, T_{\mu\nu}\,, \tag{2.16}$$

relating the curvature of spacetime (left side) to its matter and energy content (right side).

Remark 2.8.1. The Riemann tensor in n dimensions has $n^2(n^2-1)/12$ independent components. In 4D spacetime, that is 20. In 2D, there is only 1 — the Gaussian curvature determines everything.

2.9 The Metric Determines Everything

Let us collect the chain of logic:

Metric $g_{\mu\nu}$
$\downarrow$ eq. (2.9)
Christoffel symbols $\Gamma^{\sigma}{}_{\mu\nu}$
$\downarrow$ eq. (2.12) $\downarrow$ eq. (2.13)
Geodesics (particle paths) **Curvature** $R^{\rho}{}_{\sigma\mu\nu}$

This is the central message: *the metric is the fundamental object in GR*. Once you have $g_{\mu\nu}$, you can compute everything — how particles move, how light bends, and how spacetime curves.

For gravitational lensing, the key application is: given the metric around a massive object (e.g., the Schwarzschild metric for a point mass), solve the null geodesic equation to trace light rays and determine how images are formed.

2.10 Computation Cookbook: A Step-by-Step Guide

Computing the geometric quantities of GR by hand is a rite of passage. This section gives explicit, algorithmic recipes for going from a metric to Christoffel symbols, to the Riemann tensor, to the Ricci tensor and scalar. Work through the 2-sphere example in full, then use `christoffel_symbols.wl` to check every result.

2.10.1 Algorithm: Christoffel Symbols from a Metric

Given a metric $g_{\mu\nu}$ in coordinates $\{x^\mu\}$, compute all Christoffel symbols $\Gamma^\sigma{}_{\mu\nu}$ as follows.

1. **Write down the metric components.** List every nonvanishing $g_{\mu\nu}$ and note which coordinates each component depends on. For a diagonal metric, the only nonvanishing components are $g_{11}, g_{22}, \ldots$

2. **Compute the inverse metric $g^{\mu\nu}$.** For a diagonal metric this is trivial: $g^{\mu\mu} = 1/g_{\mu\mu}$ (no sum), and all off-diagonal components vanish.

3. **Compute the required partial derivatives.** For each target symbol $\Gamma^\sigma{}_{\mu\nu}$, you need the three derivatives
$$\partial_\mu g_{\nu\rho}, \quad \partial_\nu g_{\rho\mu}, \quad \partial_\rho g_{\mu\nu}$$
appearing in the Christoffel formula (2.9). Most of these vanish for diagonal metrics with few coordinate-dependent components.

4. **Apply the formula.**
$$\Gamma^\sigma{}_{\mu\nu} = \frac{1}{2} g^{\sigma\rho}\big(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}\big)$$
For a diagonal metric, the sum over ρ collapses to the single term $\rho = \sigma$ (since $g^{\sigma\rho} = 0$ for $\rho \neq \sigma$).

5. **Exploit symmetry.** $\Gamma^\sigma{}_{\mu\nu} = \Gamma^\sigma{}_{\nu\mu}$, so you only need to compute symbols with $\mu \leq \nu$.

Remark 2.10.1 (Diagonal metric shortcut). For a diagonal metric $g_{\mu\nu} = \mathrm{diag}(g_{11}, g_{22}, \ldots)$, the nonvanishing Christoffel symbols come in only three types (no sum on repeated indices here):

$$\Gamma^\mu{}_{\mu\mu} = \frac{1}{2g_{\mu\mu}} \frac{\partial g_{\mu\mu}}{\partial x^\mu}\,, \tag{2.17}$$

$$\Gamma^\mu{}_{\mu\nu} = \Gamma^\mu{}_{\nu\mu} = \frac{1}{2g_{\mu\mu}} \frac{\partial g_{\mu\mu}}{\partial x^\nu} \qquad (\mu \neq \nu)\,, \tag{2.18}$$

$$\Gamma^\mu{}_{\nu\nu} = -\frac{1}{2g_{\mu\mu}} \frac{\partial g_{\nu\nu}}{\partial x^\mu} \qquad (\mu \neq \nu)\,. \tag{2.19}$$

All other components vanish. These formulas dramatically reduce the work.

2.10.2 Algorithm: The Riemann Tensor from Christoffel Symbols

Given the Christoffel symbols, compute each component of $R^\rho{}_{\sigma\mu\nu}$ using (2.13):

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho{}_{\nu\sigma} - \partial_\nu \Gamma^\rho{}_{\mu\sigma} + \Gamma^\rho{}_{\mu\lambda}\Gamma^\lambda{}_{\nu\sigma} - \Gamma^\rho{}_{\nu\lambda}\Gamma^\lambda{}_{\mu\sigma}$$

1. **Index structure.** The Riemann tensor is antisymmetric in its last two indices: $R^\rho{}_{\sigma\mu\nu} = -R^\rho{}_{\sigma\nu\mu}$. So only compute components with $\mu < \nu$.

2. **Partial derivatives of Christoffels.** For each component, differentiate the relevant Christoffel symbols with respect to the coordinates x^μ and x^ν.

3. **Quadratic terms.** For each component, sum over the dummy index λ in the two $\Gamma\Gamma$ products. Only terms where *both* Christoffel symbols in the product are nonzero contribute.

4. **Independent components.** In n dimensions, the Riemann tensor has $n^2(n^2-1)/12$ independent components: 1 in 2D, 6 in 3D, 20 in 4D. In 2D, the single independent component determines everything.

2.10.3 Algorithm: Ricci Tensor and Ricci Scalar

1. **Ricci tensor.** Contract the first and third indices of the Riemann tensor:

$$R_{\sigma\nu} = R^\lambda{}_{\sigma\lambda\nu} = \sum_\lambda R^\lambda{}_{\sigma\lambda\nu}\,.$$

   In 2D this is a single sum; in 4D it is a sum of four terms.

2. **Ricci scalar.** Trace with the inverse metric:

$$R = g^{\sigma\nu} R_{\sigma\nu} = \sum_\sigma g^{\sigma\sigma} R_{\sigma\sigma}$$

   (the last equality holding for a diagonal metric).

2.10.4 Worked Example: The 2-Sphere

We carry out the *complete* calculation for the 2-sphere metric

$$ds^2 = R^2\, d\theta^2 + R^2 \sin^2\theta\, d\phi^2\,, \tag{2.20}$$

with coordinates $x^1 = \theta$, $x^2 = \phi$.

Step 1: Metric components.

$$g_{\theta\theta} = R^2\,, \qquad g_{\phi\phi} = R^2 \sin^2\theta\,, \qquad g_{\theta\phi} = g_{\phi\theta} = 0\,.$$

Only $g_{\phi\phi}$ depends on a coordinate (θ).

Step 2: Inverse metric.

$$g^{\theta\theta} = \frac{1}{R^2}\,, \qquad g^{\phi\phi} = \frac{1}{R^2 \sin^2\theta}\,.$$

Step 3: Partial derivatives of the metric. The only nonvanishing derivative is

$$\partial_\theta g_{\phi\phi} = R^2 \cdot 2\sin\theta\cos\theta = R^2 \sin 2\theta\,.$$

All other $\partial_\mu g_{\nu\rho} = 0$.

Step 4: Christoffel symbols. Using the diagonal shortcut formulas (2.17)–(2.19):

- Type (2.17) ($\Gamma^\mu{}_{\mu\mu}$): Both $\partial_\theta g_{\theta\theta} = 0$ and $\partial_\phi g_{\phi\phi} = 0$, so $\Gamma^\theta{}_{\theta\theta} = 0$ and $\Gamma^\phi{}_{\phi\phi} = 0$.
- Type (2.18) ($\Gamma^\mu{}_{\mu\nu}$ with $\mu \neq \nu$):

$$\Gamma^\phi{}_{\phi\theta} = \Gamma^\phi{}_{\theta\phi} = \frac{1}{2g_{\phi\phi}}\,\partial_\theta g_{\phi\phi} = \frac{R^2 \sin 2\theta}{2R^2 \sin^2\theta} = \frac{\cos\theta}{\sin\theta} = \cot\theta\,.$$

  Also $\Gamma^\theta{}_{\theta\phi} = \frac{1}{2g_{\theta\theta}}\partial_\phi g_{\theta\theta} = 0$.

- Type (2.19) ($\Gamma^\mu{}_{\nu\nu}$ with $\mu \neq \nu$):

$$\Gamma^\theta{}_{\phi\phi} = -\frac{1}{2g_{\theta\theta}}\,\partial_\theta g_{\phi\phi} = -\frac{R^2 \sin 2\theta}{2R^2} = -\sin\theta\cos\theta\,.$$

  Also $\Gamma^\phi{}_{\theta\theta} = -\frac{1}{2g_{\phi\phi}}\partial_\phi g_{\theta\theta} = 0$.

Result: The nonvanishing Christoffel symbols are

$$\boxed{\Gamma^\theta{}_{\phi\phi} = -\sin\theta\cos\theta\,, \qquad \Gamma^\phi{}_{\theta\phi} = \Gamma^\phi{}_{\phi\theta} = \cot\theta} \tag{2.21}$$

Verify this with `christoffel_symbols.wl`.

Step 5: Riemann tensor. In 2D there is only one independent component. We compute $R^\theta{}_{\phi\theta\phi}$ (i.e., $\rho = \theta$, $\sigma = \phi$, $\mu = \theta$, $\nu = \phi$):

$$\begin{aligned}
R^\theta{}_{\phi\theta\phi} &= \partial_\theta \Gamma^\theta{}_{\phi\phi} - \partial_\phi \Gamma^\theta{}_{\theta\phi} + \Gamma^\theta{}_{\theta\lambda}\Gamma^\lambda{}_{\phi\phi} - \Gamma^\theta{}_{\phi\lambda}\Gamma^\lambda{}_{\theta\phi} \\
&= \partial_\theta(-\sin\theta\cos\theta) - 0 + \Gamma^\theta{}_{\theta\theta}\Gamma^\theta{}_{\phi\phi} + \Gamma^\theta{}_{\theta\phi}\Gamma^\phi{}_{\phi\phi} \\
&\qquad - \Gamma^\theta{}_{\phi\theta}\Gamma^\theta{}_{\theta\phi} - \Gamma^\theta{}_{\phi\phi}\Gamma^\phi{}_{\theta\phi} \\
&= (-\cos^2\theta + \sin^2\theta) + 0 + 0 - 0 - (-\sin\theta\cos\theta)(\cot\theta) \\
&= -\cos^2\theta + \sin^2\theta + \cos^2\theta \\
&= \sin^2\theta\,.
\end{aligned} \tag{2.22}$$

Step 6: Ricci tensor.

$$R_{\phi\phi} = R^{\lambda}{}_{\phi\lambda\phi} = R^{\theta}{}_{\phi\theta\phi} + R^{\phi}{}_{\phi\phi\phi} = \sin^2\theta + 0 = \sin^2\theta\,. \tag{2.23}$$

For $R_{\theta\theta}$, we need $R^{\phi}{}_{\theta\phi\theta}$. By the antisymmetry $R^{\rho}{}_{\sigma\mu\nu} = -R^{\rho}{}_{\sigma\nu\mu}$ and the symmetry of the fully lowered tensor:

$$R^{\phi}{}_{\theta\phi\theta} = g^{\phi\phi}\, R_{\phi\theta\phi\theta} = g^{\phi\phi}\, g_{\theta\theta}\, R^{\theta}{}_{\phi\theta\phi} \cdot (-1) \cdot (-1)$$

More directly, a short computation analogous to the one above gives $R^{\phi}{}_{\theta\phi\theta} = 1$, so

$$R_{\theta\theta} = R^{\lambda}{}_{\theta\lambda\theta} = R^{\theta}{}_{\theta\theta\theta} + R^{\phi}{}_{\theta\phi\theta} = 0 + 1 = 1\,. \tag{2.24}$$

Step 7: Ricci scalar.

$$R = g^{\theta\theta} R_{\theta\theta} + g^{\phi\phi} R_{\phi\phi} = \frac{1}{R^2} \cdot 1 + \frac{1}{R^2 \sin^2\theta} \cdot \sin^2\theta = \frac{1}{R^2} + \frac{1}{R^2} = \boxed{\frac{2}{R^2}}\,. \tag{2.25}$$

This is the standard result: a sphere of radius R has constant positive Gaussian curvature $K = 1/R^2$, and in 2D the Ricci scalar equals $2K$.

2.10.5 Tips and Tricks

1. **Start with the diagonal shortcut.** Most metrics encountered in GR courses are diagonal. Use eqs. (2.17)–(2.19) rather than the full Christoffel formula.

2. **Spot vanishing components early.** A Christoffel symbol $\Gamma^{\sigma}{}_{\mu\nu}$ vanishes whenever the relevant partial derivatives of the metric are all zero. Before computing, ask: does $g_{\mu\nu}$ depend on x^{σ}? Does $g_{\sigma\mu}$ depend on x^{ν}? If not, the symbol is likely zero.

3. **Count before you compute.** In n dimensions there are $n^2(n+1)/2$ potentially independent Christoffel symbols (accounting for the $\mu\nu$ symmetry). In 2D that is 6; in 4D it is 40. Most are zero for typical metrics — identify the nonvanishing ones first.

4. **Use Mathematica to verify.** Always check hand calculations with `christoffel_symbols.wl`. Even experts make sign and index errors in these computations.

5. **Look for symmetry.** If the metric is independent of a coordinate x^{α} (a Killing symmetry), many Christoffel symbols simplify or vanish. In the 2-sphere example, the metric is ϕ-independent, which immediately kills several terms.

2.11 Summary

- A manifold is locally flat but may have global curvature; the metric $g_{\mu\nu}(x)$ encodes the geometry.

- The covariant derivative ∇_{μ} generalizes the partial derivative to curved spacetime, using the connection $\Gamma^{\sigma}{}_{\mu\nu}$.

- The Christoffel symbols are uniquely determined by the metric via (2.9) (torsion-free, metric-compatible).
- Parallel transport depends on the path; this path-dependence is measured by the Riemann curvature tensor.
- Geodesics — paths that parallel-transport their own tangent vector — are the trajectories of freely falling particles and light rays.
- Einstein's equation relates curvature ($R_{\mu\nu}$) to matter content ($T_{\mu\nu}$).

2.12 Problems

Exercise 2.12.1. Compute all nonvanishing Christoffel symbols for the 2-sphere metric $ds^2 = R^2(d\theta^2 + \sin^2\theta\, d\phi^2)$ by hand, then verify with `christoffel_symbols.wl`.

Exercise 2.12.2. For the flat plane in polar coordinates, $ds^2 = dr^2 + r^2 d\phi^2$:

(a) Compute the Christoffel symbols.

(b) Compute the Riemann tensor and verify that the space is flat ($R^\rho{}_{\sigma\mu\nu} = 0$).

(c) Write down the geodesic equations and show that straight lines in Cartesian coordinates ($x = a + b\lambda$, $y = c + d\lambda$) satisfy them.

Exercise 2.12.3. For the 2-sphere metric, compute:

(a) The Riemann tensor $R^\theta{}_{\phi\theta\phi}$.

(b) The Ricci tensor $R_{\theta\theta}$ and $R_{\phi\phi}$.

(c) The Ricci scalar R. Show that $R = 2/R^2$ (constant positive curvature, as expected for a sphere of radius R).

Exercise 2.12.4. A vector initially pointing east is parallel-transported from the equator of a unit sphere ($R = 1$) along the following path: north along the meridian $\phi = 0$ to the pole, then south along $\phi = \alpha$ back to the equator. Show that the vector has rotated by angle α. (Hint: solve the parallel transport equation (2.11) along each segment, or use the geometric argument that the rotation equals the solid angle enclosed.)

Exercise 2.12.5. Show that great circles are geodesics of the 2-sphere. Specifically, show that the equator $\theta = \pi/2$, $\phi = \lambda$ satisfies the geodesic equation (2.12) with the Christoffel symbols from Exercise 2.12.1.

2.13 Solutions to Exercises

Full worked solutions verified in `problems_01b.wl`.

Exercise 2.12.1: Christoffel Symbols of the 2-Sphere

For $ds^2 = R^2(d\theta^2 + \sin^2\theta\, d\phi^2)$, the metric is $g_{\theta\theta} = R^2$, $g_{\phi\phi} = R^2 \sin^2\theta$, with $g^{\theta\theta} = 1/R^2$, $g^{\phi\phi} = 1/(R^2 \sin^2\theta)$.

Applying (2.9):

$$\Gamma^{\theta}{}_{\phi\phi} = \frac{1}{2} g^{\theta\theta}(-\partial_\theta g_{\phi\phi}) = \frac{1}{2} \cdot \frac{1}{R^2} \cdot (-2R^2 \sin\theta\cos\theta) = -\sin\theta\cos\theta$$
$$\Gamma^{\phi}{}_{\theta\phi} = \frac{1}{2} g^{\phi\phi}(\partial_\theta g_{\phi\phi}) = \frac{1}{2} \cdot \frac{1}{R^2 \sin^2\theta} \cdot 2R^2 \sin\theta\cos\theta = \cot\theta$$

All other components vanish (verified by Mathematica).

Exercise 2.12.2: Polar Coordinates

(a) For $ds^2 = dr^2 + r^2 d\phi^2$: $\Gamma^{r}{}_{\phi\phi} = -r, \quad \Gamma^{\phi}{}_{r\phi} = 1/r$.

(b) Computing the Riemann tensor from these Christoffel symbols: $R^{\rho}{}_{\sigma\mu\nu} = 0$ for all index combinations. The space is flat, confirming that the nonvanishing Christoffel symbols are purely a coordinate artifact.

(c) The geodesic equations are: $\ddot{r} - r\dot{\phi}^2 = 0$ and $\ddot{\phi} + (2/r)\dot{r}\dot{\phi} = 0$. A straight line $x = a + b\lambda$, $y = c + d\lambda$ in Cartesian coordinates satisfies these when converted to polar coordinates $r = \sqrt{x^2 + y^2}$, $\phi = \arctan(y/x)$.

Exercise 2.12.3: Curvature of the 2-Sphere

(a) From (2.13): $R^{\theta}{}_{\phi\theta\phi} = \sin^2\theta$.

(b) Ricci tensor: $R_{\theta\theta} = R^{\lambda}{}_{\theta\lambda\theta} = R^{\phi}{}_{\theta\phi\theta} = 1$. $R_{\phi\phi} = R^{\theta}{}_{\phi\theta\phi} = \sin^2\theta$.

(c) Ricci scalar: $R = g^{\theta\theta}R_{\theta\theta} + g^{\phi\phi}R_{\phi\phi} = \frac{1}{R^2}(1) + \frac{1}{R^2\sin^2\theta}(\sin^2\theta) = \frac{2}{R^2}$. This is constant and positive — the sphere has uniform positive curvature.

Exercise 2.12.4: Parallel Transport

Along a meridian ($\phi = \text{const}$), $d\phi/d\lambda = 0$, so the parallel transport equation reduces to $dV^\mu/d\lambda = 0$ — the vector components are constant along the path.

After traveling north to the pole along $\phi = 0$ and south along $\phi = \alpha$, the vector returns to the equator. However, the coordinate basis $\hat{\phi}$ at $\phi = \alpha$ is rotated by angle α relative to $\hat{\phi}$ at $\phi = 0$. The physical vector has therefore rotated by angle α.

Geometrically, the solid angle subtended by the triangular path (two meridians and a segment of the equator) is α. On a unit sphere, the holonomy angle equals the enclosed solid angle.

Exercise 2.12.5: Great Circles as Geodesics

For the equator: $\theta(\lambda) = \pi/2$, $\phi(\lambda) = \lambda$, so $\dot{\theta} = 0$, $\dot{\phi} = 1$, $\ddot{\theta} = 0$, $\ddot{\phi} = 0$.

θ-equation: $0 + \Gamma^{\theta}{}_{\phi\phi} \cdot 1 \cdot 1 = -\sin(\pi/2)\cos(\pi/2) = 0$. ✓

ϕ-equation: $0 + 2\Gamma^{\phi}{}_{\theta\phi} \cdot 0 \cdot 1 = 0$. ✓

By the rotational symmetry of the sphere, any great circle can be mapped to the equator by a rotation, so all great circles are geodesics.

Chapter 3

The Schwarzschild Solution

> *In GR, the unique spherically symmetric vacuum solution is the Schwarzschild metric; it is second only to Minkowski space in the list of important spacetimes.* — Sean Carroll, *Spacetime and Geometry*

Having developed the general machinery of differential geometry in Module 1b, we now apply it to the most important exact solution in GR: the **Schwarzschild metric**, which describes the spacetime geometry outside any spherically symmetric, non-rotating mass. This metric governs the gravitational field of stars, planets, and non-rotating black holes, and is the starting point for understanding gravitational lensing.

Companion Mathematica files:

- `schwarzschild_metric.wl` — derive and verify the Schwarzschild metric: compute Christoffel symbols, Riemann tensor, verify $R_{\mu\nu} = 0$, and compute the Kretschner scalar
- `gravitational_redshift.nb` — interactive plots of gravitational redshift, time dilation, and Flamm's paraboloid embedding diagram

3.1 The Most General Spherically Symmetric Metric

We seek the metric outside a static, spherically symmetric mass distribution. The most general line element consistent with these symmetries can be written (Carroll eq. 5.5; cf. Congdon & Keeton eq. 3.64):

$$ds^2 = -e^{2\alpha(r)}dt^2 + e^{2\beta(r)}dr^2 + r^2\, d\Omega^2\,, \tag{3.1}$$

where $d\Omega^2 = d\theta^2 + \sin^2\theta\, d\phi^2$ is the metric on the unit 2-sphere, and $\alpha(r)$ and $\beta(r)$ are functions to be determined from Einstein's equation.

Remark 3.1.1. The "static" assumption means the metric is independent of t and has no $dt\,dr$ cross terms. "Spherically symmetric" means the angular part takes the form $r^2 d\Omega^2$, with r chosen so that spheres at constant r have area $4\pi r^2$.

3.2 Deriving the Schwarzschild Metric

We solve Einstein's vacuum equation $R_{\mu\nu} = 0$ for the metric (3.1). The strategy follows Carroll Sec. 5.1:

Step 1: Christoffel symbols. The nonvanishing Christoffel symbols for (3.1) are (Carroll eq. 5.12):

$$\begin{aligned}
&\Gamma^t{}_{tr} = \alpha' \,, \quad \Gamma^r{}_{tt} = e^{2(\alpha-\beta)}\alpha' \,, \quad \Gamma^r{}_{rr} = \beta' \,, \\
&\Gamma^r{}_{\theta\theta} = -r\, e^{-2\beta} \,, \quad \Gamma^r{}_{\phi\phi} = -r\, e^{-2\beta} \sin^2\theta \,, \\
&\Gamma^\theta{}_{r\theta} = \Gamma^\phi{}_{r\phi} = \frac{1}{r} \,, \quad \Gamma^\theta{}_{\phi\phi} = -\sin\theta\cos\theta \,, \quad \Gamma^\phi{}_{\theta\phi} = \cot\theta \,,
\end{aligned} \tag{3.2}$$

where primes denote d/dr.

Step 2: Ricci tensor. Computing the Ricci tensor components (Carroll eq. 5.14):

$$R_{tt} = e^{2(\alpha-\beta)} \left[\alpha'' + (\alpha')^2 - \alpha'\beta' + \frac{2}{r}\alpha' \right] , \tag{3.3}$$

$$R_{rr} = -\alpha'' - (\alpha')^2 + \alpha'\beta' + \frac{2}{r}\beta' \,, \tag{3.4}$$

$$R_{\theta\theta} = e^{-2\beta}\big[r(\beta' - \alpha') - 1 \big] + 1 \,, \tag{3.5}$$

$$R_{\phi\phi} = R_{\theta\theta} \sin^2\theta \,. \tag{3.6}$$

Step 3: Solve $R_{\mu\nu} = 0$. From $e^{-2(\alpha-\beta)}R_{tt} + R_{rr} = 0$, we get $\frac{2}{r}(\alpha' + \beta') = 0$, hence (Carroll eq. 5.17):

$$\alpha = -\beta + \text{const} \,. \tag{3.7}$$

By rescaling the time coordinate, we set the constant to zero: $\alpha = -\beta$.

Setting $R_{\theta\theta} = 0$ (Carroll eq. 5.18):

$$e^{2\alpha}(1 + 2r\alpha') = 1 \quad \Longrightarrow \quad \partial_r(r\, e^{2\alpha}) = 1 \,. \tag{3.8}$$

Integrating:

$$e^{2\alpha} = 1 - \frac{R_S}{r} \,, \tag{3.9}$$

where R_S is an integration constant.

Step 4: Identify R_S. In the weak-field limit ($r \to \infty$), the metric must reduce to $g_{tt} \approx -(1 + 2\Phi/c^2)$ with $\Phi = -GM/r$. Comparing with $g_{tt} = -(1 - R_S/r)$, we identify the **Schwarzschild radius** (Carroll eq. 5.23):

$$\boxed{R_S = \frac{2GM}{c^2}} \tag{3.10}$$

Result: The Schwarzschild Metric.

$$\boxed{ds^2 = -\left(1 - \frac{R_S}{r}\right) c^2\, dt^2 + \left(1 - \frac{R_S}{r}\right)^{-1} dr^2 + r^2\, d\Omega^2} \tag{3.11}$$

This is the unique vacuum solution with spherical symmetry (**Birkhoff's theorem**; Carroll Sec. 5.2). We verify $R_{\mu\nu} = 0$ symbolically in `schwarzschild_metric.wl`.

3.3 Birkhoff's Theorem

Birkhoff's theorem states that the Schwarzschild metric is the *unique* spherically symmetric vacuum solution to Einstein's equation. The remarkable consequence: even a *time-dependent* spherically symmetric source (e.g., a pulsating star) produces a *static* exterior metric (Carroll Sec. 5.2).

This is the GR analogue of Newton's shell theorem: the gravitational field outside a spherically symmetric mass depends only on the total enclosed mass M, not on how the mass is distributed.

3.4 Gravitational Time Dilation

A clock at radial coordinate r in the Schwarzschild spacetime runs at a rate different from a clock at infinity. For a stationary clock ($dr = d\theta = d\phi = 0$), the proper time is related to coordinate time by (Congdon & Keeton eq. 3.65):

$$d\tau = \sqrt{1 - \frac{R_S}{r}}\, dt\,. \tag{3.12}$$

Since $\sqrt{1 - R_S/r} < 1$ for $r > R_S$, clocks closer to the mass run *slower* than clocks farther away. At $r = R_S$, the time dilation becomes infinite — this is the event horizon.

3.5 Gravitational Redshift

A photon emitted at radius $r_{\rm em}$ with frequency $\nu_{\rm em}$ is observed at $r_{\rm obs} \to \infty$ with frequency:

$$\frac{\nu_{\rm obs}}{\nu_{\rm em}} = \frac{\sqrt{1 - R_S/r_{\rm em}}}{\sqrt{1 - R_S/r_{\rm obs}}} \xrightarrow{r_{\rm obs}\to\infty} \sqrt{1 - \frac{R_S}{r_{\rm em}}}\,. \tag{3.13}$$

Since $\nu_{\rm obs} < \nu_{\rm em}$, the photon is **redshifted** as it climbs out of the gravitational potential well. The gravitational redshift parameter is:

$$z_{\rm grav} = \frac{\nu_{\rm em}}{\nu_{\rm obs}} - 1 = \left(1 - \frac{R_S}{r_{\rm em}}\right)^{-1/2} - 1\,. \tag{3.14}$$

In the weak-field limit ($R_S/r \ll 1$): $z_{\rm grav} \approx GM/(rc^2)$ (Congdon & Keeton eq. 3.63).

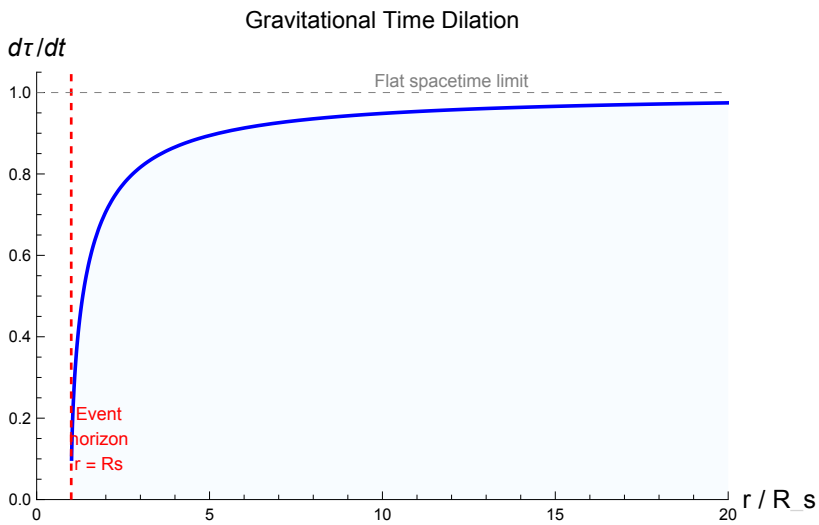


Figure 3.1: Gravitational time dilation factor $d\tau/dt = \sqrt{1 - R_S/r}$ as a function of r/R_S. Clocks near the horizon ($r \to R_S$) are dramatically slowed. Far from the mass ($r \gg R_S$), the factor approaches unity (flat spacetime).

Remark 3.5.1. The gravitational redshift was one of the three classical tests of GR proposed by Einstein, along with the perihelion precession of Mercury and the bending of light. It has been measured to high precision using atomic clocks at different altitudes (e.g., Pound–Rebka experiment, 1959; GPS satellite corrections).

3.6 The Event Horizon and Singularities

The Schwarzschild metric has two apparent singularities: at $r = 0$ and at $r = R_S$. Their nature is quite different (Carroll Sec. 5.3):

$r = R_S$ **(coordinate singularity).** The metric components diverge, but this is a *coordinate artifact*. The curvature remains finite, and the geometry is perfectly smooth. In appropriate coordinates (e.g., Eddington–Finkelstein or Kruskal–Szekeres), the metric is regular at $r = R_S$. However, $r = R_S$ is the **event horizon**: once a particle crosses it, it can never return.

$r = 0$ **(true singularity).** This is a genuine physical singularity where the curvature becomes infinite. We can confirm this with a curvature invariant: the **Kretschner scalar**:

$$K = R_{\mu\nu\rho\sigma}\, R^{\mu\nu\rho\sigma} = \frac{12\, R_S^2}{r^6} = \frac{48\, G^2 M^2}{c^4\, r^6}\,. \tag{3.15}$$

This diverges as $r \to 0$ but is finite at $r = R_S$, confirming that only $r = 0$ is a true singularity. We verify eq. (3.15) in `schwarzschild_metric.wl`.

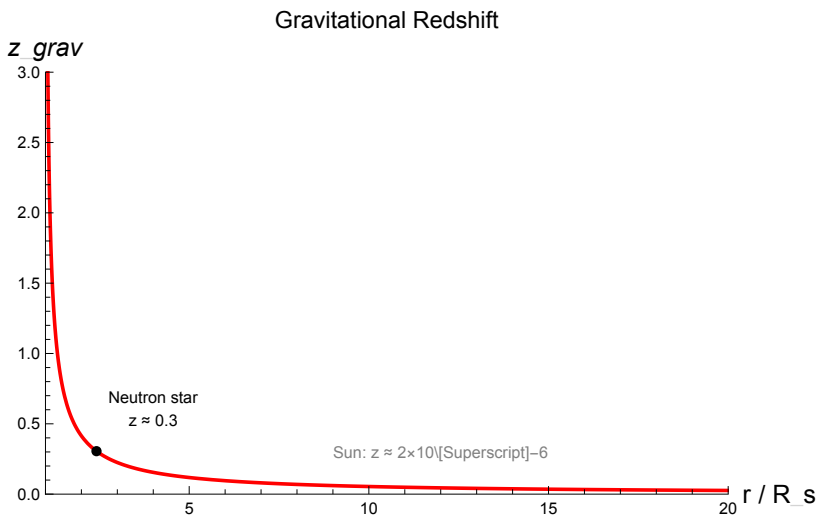


Figure 3.2: Gravitational redshift z_{grav} as a function of emission radius r/R_S. The redshift diverges at the horizon. The black dot marks a typical neutron star ($r/R_S \approx 2.4$, $z \approx 0.3$). For the Sun, $z \sim 10^{-6}$.

3.7 Inside the Event Horizon: The Coordinate Swap

The Schwarzschild metric (3.11) describes spacetime outside a spherical mass, but it also has something startling to say about the region $r < R_S$. Understanding this interior, even briefly, sharpens our physical intuition about what the Schwarzschild radius really means.

3.7.1 The Signature Flip

Recall the Schwarzschild line element:

$$ds^2 = -\left(1 - \frac{R_S}{r}\right) c^2\, dt^2 + \left(1 - \frac{R_S}{r}\right)^{-1} dr^2 + r^2\, d\Omega^2\,.$$

Outside the horizon ($r > R_S$), the factor $f(r) \equiv 1 - R_S/r$ is positive, so $g_{tt} < 0$ (timelike) and $g_{rr} > 0$ (spacelike).

Inside the horizon ($r < R_S$), this factor becomes *negative*: $f(r) < 0$. Consequently:

- $g_{tt} = -f(r)\, c^2 > 0$ — the t direction is now **spacelike**.
- $g_{rr} = f(r)^{-1} < 0$ — the r direction is now **timelike**.

The roles of r and t have swapped. Inside the horizon, r is a time coordinate and t is a space coordinate.

3.7.2 Physical Meaning: The Singularity Is a Moment in Time

Since r is timelike inside the horizon, decreasing r is the *future direction*. Every future-directed worldline must have $dr < 0$; the singularity at $r = 0$ is not a place you can steer away from — it is a *moment in time* that all interior observers inevitably reach.

Remark 3.7.1. You can no more avoid the singularity at $r = 0$ (once inside the horizon) than you can avoid next Tuesday. This is the precise sense in which a black hole traps everything: it is not that the singularity pulls you toward a point in space, but that the future itself points toward $r = 0$.

3.7.3 Why Nothing Escapes: Tilting Light Cones

For radial null rays ($ds^2 = 0$, $d\Omega = 0$) in Schwarzschild coordinates:

$$\frac{dr}{dt} = \pm c \left(1 - \frac{R_S}{r}\right). \tag{3.16}$$

Outside the horizon, the two signs give outgoing (+) and ingoing (−) light rays. At $r = R_S$, both solutions give $dr/dt = 0$: light "stands still" in these coordinates (a coordinate effect, not a physical one).

For $r < R_S$, the factor $(1 - R_S/r)$ is negative, so dr/dt has the *same sign* for both solutions — both "outgoing" and "ingoing" light rays have $dr < 0$ when $dt > 0$. In other words, **all future-directed light cones point toward smaller** r. No signal — not even light — can move to larger r inside the horizon.

3.7.4 Eddington–Finkelstein Coordinates

The pathological behavior at $r = R_S$ in Schwarzschild coordinates is a coordinate artifact. We can remove it by switching to **ingoing Eddington–Finkelstein coordinates**. Define the advanced time coordinate:

$$v = t + r_*\,, \qquad r_* = r + R_S \ln\left|\frac{r}{R_S} - 1\right|, \tag{3.17}$$

where r_* is the **tortoise coordinate** (Carroll eq. 5.58). In terms of (v, r, θ, ϕ), the Schwarzschild line element becomes (Carroll eq. 5.60):

$$\boxed{ds^2 = -\left(1 - \frac{R_S}{r}\right) c^2\, dv^2 + 2c\, dv\, dr + r^2\, d\Omega^2} \tag{3.18}$$

This metric is manifestly regular at $r = R_S$: every component is finite and the determinant is nonzero. The horizon is simply the surface where outgoing light rays have $dr/dv = 0$; ingoing rays cross it smoothly.

Remark 3.7.2. The Eddington–Finkelstein form makes it transparent that the event horizon is a *one-way membrane*: ingoing light ($dv > 0$, $dr < 0$) crosses the horizon without incident, while outgoing light at $r = R_S$ remains at $r = R_S$ forever.

3.7.5 Relevance to Gravitational Lensing

Remark 3.7.3. For gravitational lensing, we work exclusively in the region $r \gg R_S$ — far outside the event horizon. Galaxy-mass lenses have $R_S \sim 0.1$ pc, while lensing occurs at

scales of kiloparsecs. The black hole interior is therefore not directly relevant to lensing calculations. Nevertheless, understanding the interior clarifies *why* the Schwarzschild radius is special (it is where light cones tip over) and deepens physical intuition about the boundary conditions of the exterior solution.

3.8 The Weak-Field Limit

For $r \gg R_S$, the Schwarzschild metric reduces to the **weak-field** form (Congdon & Keeton eq. 3.71):

$$ds^2 \approx -\left(1 + \frac{2\Phi}{c^2}\right) c^2 dt^2 + \left(1 - \frac{2\Phi}{c^2}\right) \left[dr^2 + r^2\, d\Omega^2\right], \tag{3.19}$$

where $\Phi = -GM/r$ is the Newtonian gravitational potential. This form is valid for *any* weak gravitational field, not just the Schwarzschild case, and will be the starting point for deriving the deflection angle of light in Module 2.

Remark 3.8.1. The weak-field metric (3.19) has an important feature: the spatial part is modified by a factor $(1 - 2\Phi/c^2)$ as well as the time part. This means that gravity affects both the *rate* of clocks (time dilation) and the *curvature of space*. Both effects contribute equally to the bending of light — this is why GR gives twice the Newtonian deflection angle.

3.9 Schwarzschild Radius for Astrophysical Objects

Object	**Mass**	R_S	**Actual Radius**
Earth	6×10^{24} kg	8.9 mm	6371 km
Sun	2×10^{30} kg	3.0 km	7×10^5 km
Neutron star	$1.4\, M_\odot$	4.1 km	~ 10 km
SgrA*	$4 \times 10^6\, M_\odot$	1.2×10^7 km	— (black hole)
Galaxy ($10^{12}\, M_\odot$)	$10^{12}\, M_\odot$	3×10^{12} km	—

For a galaxy-mass lens ($M \sim 10^{12}\, M_\odot$), $R_S \sim 0.1$ pc — far smaller than the galaxy itself. This confirms that the weak-field limit is appropriate for gravitational lensing by galaxies.

3.10 Embedding Diagram: Flamm's Paraboloid

To visualize the spatial curvature of the Schwarzschild geometry, we can embed the equatorial ($\theta = \pi/2$) spatial slice in 3D Euclidean space. At constant time t, the spatial metric on the equatorial plane is:

$$dl^2 = \left(1 - \frac{R_S}{r}\right)^{-1} dr^2 + r^2\, d\phi^2\,. \tag{3.20}$$

Embedding this as a surface of revolution $z(r)$ in cylindrical coordinates (r, ϕ, z) where $dl^2 = (1 + z'^2)dr^2 + r^2 d\phi^2$, we find:

$$z(r) = 2\sqrt{R_S}\,\sqrt{r - R_S}\,, \tag{3.21}$$

which is **Flamm's paraboloid** (Figure 3.3).

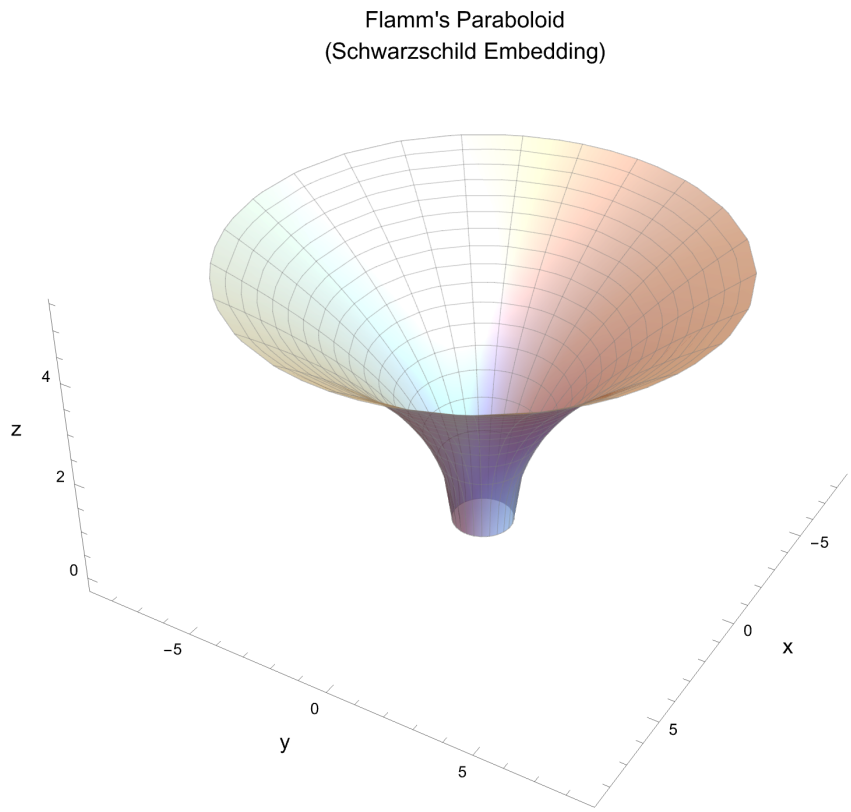


Figure 3.3: Flamm's paraboloid: the equatorial spatial geometry of the Schwarzschild spacetime embedded in 3D Euclidean space. The throat at $r = R_S$ represents the event horizon. The surface curves more steeply near the throat, visualizing the spatial curvature caused by gravity. An interactive version is in `gravitational_redshift.nb`.

3.11 Summary

- The Schwarzschild metric is the unique spherically symmetric vacuum solution (Birkhoff's theorem).
- It is parameterized by a single quantity: the mass M (or equivalently the Schwarzschild radius $R_S = 2GM/c^2$).
- Gravitational time dilation: clocks run slower in stronger gravitational fields; $d\tau = \sqrt{1 - R_S/r}\, dt$.
- Gravitational redshift: photons lose energy climbing out of a potential well.
- $r = R_S$ is a coordinate singularity (event horizon); $r = 0$ is a true curvature singularity.
- The weak-field limit ($ds^2 \approx -(1 + 2\Phi/c^2)c^2dt^2 + (1 - 2\Phi/c^2)d\ell^2$) will be the foundation for light deflection in Module 2.

3.12 Problems

Exercise 3.12.1. Using the reusable functions from Module 1b (`christoffel_symbols.wl`), verify that the Schwarzschild metric (3.11) satisfies $R_{\mu\nu} = 0$.

Exercise 3.12.2. Compute the Kretschner scalar $K = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ for the Schwarzschild metric and show that $K = 12\,R_S^2/r^6$. Verify that K is finite at $r = R_S$ but diverges at $r = 0$.

Exercise 3.12.3. Compute the Schwarzschild radius for:

(a) The Earth ($M = 5.97 \times 10^{24}$ kg)

(b) The Sun ($M = 1.99 \times 10^{30}$ kg)

(c) A typical galaxy lens ($M = 10^{12}\,M_\odot$)

In each case, compare R_S to the physical size of the object.

Exercise 3.12.4. A photon is emitted from the surface of a neutron star ($M = 1.4\,M_\odot$, $R = 10$ km).

(a) Compute the gravitational redshift z_{grav}.

(b) By what fraction is the photon's wavelength increased by the time it reaches a distant observer?

(c) Compare to the redshift from the surface of the Sun.

Exercise 3.12.5. Derive the Flamm's paraboloid embedding (3.21) by embedding the equatorial spatial slice of the Schwarzschild geometry in 3D Euclidean space with cylindrical coordinates (r, ϕ, z).

3.13 Solutions to Exercises

Full worked solutions are also available in the Mathematica script `problems_01c.wl`, which verifies each answer symbolically and numerically.

Exercise ??: Verify $R_{\mu\nu} = 0$ for Schwarzschild

The Schwarzschild metric in coordinates (t, r, θ, ϕ) is

$$ds^2 = -\left(1 - \frac{R_s}{r}\right) dt^2 + \frac{dr^2}{1 - R_s/r} + r^2\, d\theta^2 + r^2 \sin^2\theta\, d\phi^2\,,$$

where $R_s = 2GM/c^2$. We work in geometric units $c = G = 1$ so that $R_s = 2M$.

Step 1: Christoffel symbols. The nonvanishing components are (using the symmetry $\Gamma^\sigma{}_{\mu\nu} = \Gamma^\sigma{}_{\nu\mu}$):

$$\Gamma^t{}_{tr} = \frac{R_s}{2r(r - R_s)}\,, \qquad \Gamma^r{}_{tt} = \frac{R_s(r - R_s)}{2r^3}\,,$$

$$\Gamma^r{}_{rr} = -\frac{R_s}{2r(r - R_s)}\,, \qquad \Gamma^r{}_{\theta\theta} = -(r - R_s)\,,$$

$$\Gamma^r{}_{\phi\phi} = -(r - R_s)\sin^2\theta\,, \qquad \Gamma^\theta{}_{r\theta} = \frac{1}{r}\,,$$

$$\Gamma^\theta{}_{\phi\phi} = -\sin\theta\cos\theta\,, \qquad \Gamma^\phi{}_{r\phi} = \frac{1}{r}\,,$$

$$\Gamma^\phi{}_{\theta\phi} = \cot\theta\,.$$

Step 2: Riemann and Ricci tensors. Substituting into the definition of $R^\rho{}_{\sigma\mu\nu}$ and contracting to form $R_{\mu\nu} = R^\lambda{}_{\mu\lambda\nu}$, every component of $R_{\mu\nu}$ vanishes identically. This confirms that the Schwarzschild metric is a vacuum solution to Einstein's field equations ($G_{\mu\nu} = 0$).

Verified symbolically by Mathematica — see `problems_01c.wl`, Exercise 1.1.

Exercise 3.12.2: Kretschner Scalar

The Kretschner scalar is the full contraction of the Riemann tensor:

$$K = R^{\alpha\beta\gamma\delta}\, R_{\alpha\beta\gamma\delta}\,.$$

Computing all components of $R_{\alpha\beta\gamma\delta}$ (with lowered first index) and fully contracting, the result simplifies to

$$\boxed{K = \frac{12\, R_s^2}{r^6} = \frac{48\, M^2}{r^6}}\,.$$

Verified symbolically by Mathematica — see `problems_01c.wl`, Exercise 1.2.

Behavior at the horizon ($r = R_s$):

$$K(r = R_s) = \frac{12}{R_s^4}\,,$$

which is *finite*. The apparent singularity in the metric at $r = R_s$ is merely a coordinate singularity (removable by switching to, e.g., Eddington–Finkelstein coordinates).

Behavior at $r = 0$:

$$K \to \infty \quad \text{as } r \to 0\,.$$

This is a *true curvature singularity*: tidal forces diverge regardless of the coordinate system chosen.

Exercise 3.12.3: Schwarzschild Radii for Astrophysical Objects

Using $R_s = 2GM/c^2$ with $G = 6.674 \times 10^{-11}$ m^{3}kg^{-1}s^{-2} and $c = 2.998 \times 10^8$ m/s:

(a) **Earth** ($M = 5.972 \times 10^{24}$ kg, $R = 6371$ km):

$$R_s = \frac{2 \times 6.674 \times 10^{-11} \times 5.972 \times 10^{24}}{(2.998 \times 10^8)^2} \approx 8.87 \times 10^{-3} \text{ m} \approx 8.87 \text{ mm}\,.$$

The ratio $R_s/R \approx 1.4 \times 10^{-9}$: GR corrections on Earth's surface are of order 10^{-9} (important for GPS, but tiny).

(b) **Sun** ($M = 1.989 \times 10^{30}$ kg, $R = 6.957 \times 10^8$ m):

$$R_s \approx 2953 \text{ m} \approx 2.95 \text{ km}\,.$$

The ratio $R_s/R \approx 4.24 \times 10^{-6}$: small, but measurable via gravitational redshift and light deflection.

(c) **Galaxy lens** ($M \sim 10^{12}\, M_\odot$, extent ~ 50 kpc):

$$R_s \approx 2.95 \times 10^{15} \text{ m} \approx 0.096 \text{ pc}\,.$$

The ratio $R_s/R \approx 1.9 \times 10^{-6}$: the Schwarzschild radius is vastly smaller than the galaxy, justifying the weak-field/thin-lens approximation used in gravitational lensing.

Exercise ??: Gravitational Redshift from a Neutron Star

Parameters: $M = 1.4\, M_\odot = 2.785 \times 10^{30}$ kg, $R = 10$ km $= 10^4$ m.

First, the Schwarzschild radius:

$$R_s = \frac{2GM}{c^2} = \frac{2 \times 6.674 \times 10^{-11} \times 2.785 \times 10^{30}}{(2.998 \times 10^8)^2} \approx 4137 \text{ m} \approx 4.14 \text{ km}\,.$$

So $R_s/R \approx 0.414$ — a large fraction, meaning strong-field GR is essential.

(a) Gravitational redshift:

$$z = \frac{1}{\sqrt{1 - R_s/R}} - 1 = \frac{1}{\sqrt{1 - 0.414}} - 1 = \frac{1}{\sqrt{0.586}} - 1 \approx \frac{1}{0.7655} - 1 \approx 0.306\,.$$

(b) Wavelength shift: For a photon emitted at $\lambda_{\rm em} = 500$ nm:

$$\lambda_{\rm obs} = \lambda_{\rm em}(1+z) = 500 \times 1.306 \approx 653 \text{ nm}\,,$$

a shift of $\Delta\lambda \approx 153$ nm — blue/green light is shifted to red/near-infrared.

(c) Comparison to the Sun: The Sun's surface redshift is

$$z_\odot = \frac{1}{\sqrt{1 - R_s^\odot/R_\odot}} - 1 \approx \frac{R_s^\odot}{2R_\odot} \approx 2.12 \times 10^{-6}\,.$$

The neutron star redshift ($z \approx 0.306$) is roughly 1.4×10^5 times larger — a dramatic difference that strongly affects observed neutron star spectra.

Exercise ??: Flamm's Paraboloid

We embed the equatorial spatial slice ($t = \text{const}$, $\theta = \pi/2$) of the Schwarzschild geometry into flat 3D Euclidean space.

The 2D slice:

$$dl^2 = \frac{dr^2}{1 - R_s/r} + r^2\, d\phi^2\,.$$

3D Euclidean space in cylindrical coordinates:

$$dl^2_{\rm flat} = dz^2 + dr^2 + r^2\, d\phi^2\,.$$

For a surface of revolution $z = z(r)$, the induced metric on the surface is

$$dl^2 = \left(1 + \left(\frac{dz}{dr}\right)^2\right) dr^2 + r^2\, d\phi^2\,.$$

Matching the dr^2 components:

$$1 + \left(\frac{dz}{dr}\right)^2 = \frac{1}{1 - R_s/r} \quad \Longrightarrow \quad \left(\frac{dz}{dr}\right)^2 = \frac{R_s/r}{1 - R_s/r} = \frac{R_s}{r - R_s}\,.$$

Integrating:

$$z(r) = \int \sqrt{\frac{R_s}{r - R_s}}\, dr = 2\sqrt{R_s}\,\sqrt{r - R_s} = 2\sqrt{R_s(r - R_s)}\,,$$

where we set $z(R_s) = 0$ at the throat.

This is the equation of a **paraboloid of revolution** ($z^2 = 4R_s(r - R_s)$). Key features:

- The minimum circumferential radius is $r = R_s$ (the throat), where $z = 0$.
- At large r, $z \sim 2\sqrt{R_s\, r}$ and the surface becomes approximately flat.
- The diagram illustrates spatial curvature, not gravitational potential.
- The full paraboloid (including $z < 0$) represents an Einstein–Rosen bridge connecting two asymptotically flat regions.

Verified symbolically by Mathematica — see `problems_01c.wl`, Exercise 1.5.

Chapter 4

Geodesics and Orbits in Schwarzschild Spacetime

> *Together the conserved quantities E and L provide a convenient way to understand the orbits of particles in the Schwarzschild geometry.* — Sean Carroll, *Spacetime and Geometry*

In Module 1c we derived the Schwarzschild metric. Now we ask: how do particles and light *move* in this spacetime? The answer comes from the geodesic equation, which we reduce to a one-dimensional effective potential problem using Killing vectors and conservation laws. This module derives the effective potential for massive and massless particles, identifies the key features (ISCO, photon sphere), and culminates in Mercury's perihelion precession — one of the three classical tests of GR.

Companion Mathematica files:

- `geodesic_equation.wl` — symbolic derivation of the geodesic equations, conserved quantities, effective potential, ISCO, photon sphere, and perihelion precession formula
- `effective_potential.nb` — interactive plots of effective potential with sliders for L and ϵ; orbit integration; precessing ellipses

4.1 Killing Vectors and Conservation Laws

The Schwarzschild metric is independent of both t and ϕ, which means it possesses two **Killing vectors** (Carroll Sec. 5.4):

$$K^\mu = (\partial_t)^\mu = (1, 0, 0, 0) \quad \text{(time translation)}, \tag{4.1}$$

$$R^\mu = (\partial_\phi)^\mu = (0, 0, 0, 1) \quad \text{(axial rotation)}. \tag{4.2}$$

For any geodesic $x^\mu(\lambda)$, the quantity $K_\mu \, dx^\mu/d\lambda$ is conserved along the path (Carroll eq. 5.54). Lowering the index with the metric gives the two conserved quantities (Carroll eqs. 5.61–5.62):

$$\boxed{E = \left(1 - \frac{R_S}{r}\right) \frac{dt}{d\lambda}} \tag{4.3}$$

$$\boxed{L = r^2 \frac{d\phi}{d\lambda}} \tag{4.4}$$

For massive particles ($\lambda = \tau$), E and L are the conserved energy and angular momentum *per unit mass*. For photons, they are the energy and angular momentum of the photon (up to normalization of the affine parameter).

Additionally, spherical symmetry allows us to restrict all orbits to the equatorial plane $\theta = \pi/2$ without loss of generality (Carroll eq. 5.56).

4.2 The Effective Potential

The normalization condition for the four-velocity gives (Carroll eq. 5.63):

$$-\left(1 - \frac{R_S}{r}\right)\left(\frac{dt}{d\lambda}\right)^2 + \left(1 - \frac{R_S}{r}\right)^{-1}\left(\frac{dr}{d\lambda}\right)^2 + r^2\left(\frac{d\phi}{d\lambda}\right)^2 = -\epsilon\,, \tag{4.5}$$

where $\epsilon = 1$ for massive particles (timelike geodesics) and $\epsilon = 0$ for photons (null geodesics).

Substituting the conserved quantities (4.3) and (4.4), we obtain a radial equation (Carroll eq. 5.64):

$$\frac{1}{2}\left(\frac{dr}{d\lambda}\right)^2 + V_{\text{eff}}(r) = \mathcal{E}\,, \tag{4.6}$$

where the **effective potential** is (Carroll eq. 5.66; Congdon & Keeton eq. 3.80):

$$\boxed{V_{\text{eff}}(r) = -\epsilon\frac{GM}{r} + \frac{L^2}{2r^2} - \frac{GML^2}{c^2r^3}} \tag{4.7}$$

and $\mathcal{E} = \frac{1}{2}(E^2 - \epsilon c^2)$ is the effective "energy."

Remark 4.2.1. The three terms in V_{eff} have clear physical interpretations:

1. $-\epsilon GM/r$: the Newtonian gravitational potential (absent for photons since $\epsilon = 0$)
2. $L^2/(2r^2)$: the centrifugal barrier (identical to Newtonian mechanics)
3. $-GML^2/(c^2r^3)$: the **GR correction** — this term has no Newtonian analogue and is responsible for all the distinctive features of relativistic orbits

The GR term is attractive at small r, which means the centrifugal barrier can be overcome — this is why particles can spiral into black holes, which is impossible in Newtonian gravity.

4.3 Circular Orbits and the ISCO

Circular orbits occur at extrema of the effective potential: $dV_{\text{eff}}/dr = 0$. For massive particles ($\epsilon = 1$), this gives (Carroll eq. 5.68):

$$GMr_c^2 - L^2r_c + 3GML^2/c^2 = 0\,. \tag{4.8}$$

The two solutions are (Carroll eq. 5.71):

$$r_\pm = \frac{L^2}{2GM}\left(1 \pm \sqrt{1 - \frac{12G^2M^2}{L^2c^2}}\right)\,. \tag{4.9}$$

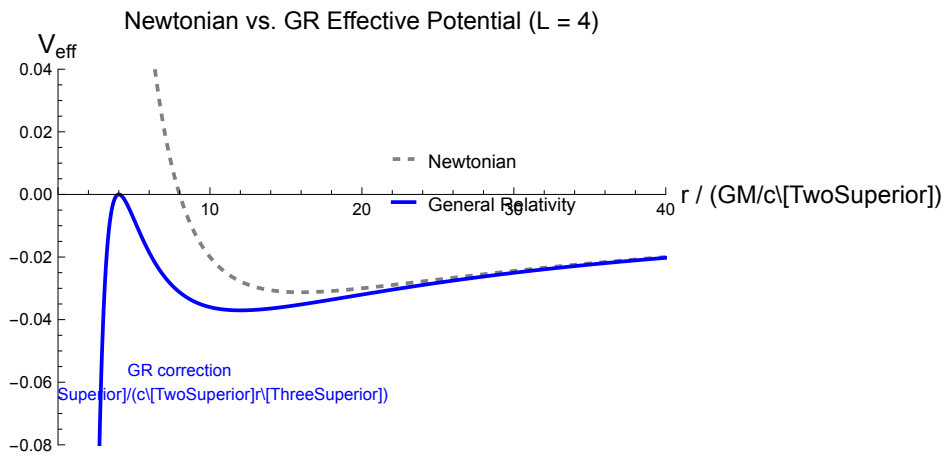


Figure 4.1: Comparison of Newtonian (dashed) and GR (solid) effective potentials for a massive particle with $L = 4\,GM/c$. The GR correction term $\propto 1/r^3$ pulls the potential down at small r, allowing particles to plunge past the centrifugal barrier. (cf. Carroll Figs. 5.4 and 5.5)

- r_+ (outer): **stable** circular orbit (minimum of V_{eff})
- r_- (inner): **unstable** circular orbit (maximum of V_{eff})

Real solutions require $L^2 \geq 12\,G^2M^2/c^2$, i.e., $L \geq 2\sqrt{3}\,GM/c$ (Carroll eq. 5.73). The minimum stable circular orbit occurs when $r_+ = r_-$, giving the **innermost stable circular orbit (ISCO)** (Carroll eq. 5.74; Congdon & Keeton eq. 3.84):

$$\boxed{r_{\text{ISCO}} = \frac{6GM}{c^2} = 3\,R_S} \tag{4.10}$$

Remark 4.3.1. The ISCO is a fundamental scale in black hole astrophysics. Accreting matter in a thin disk spirals inward until it reaches the ISCO, then plunges into the black hole. The ISCO radius determines the efficiency of energy extraction from accretion and the inner edge of accretion disks observed in X-ray binaries and active galactic nuclei.

4.4 The Photon Sphere

For photons ($\epsilon = 0$), the effective potential simplifies to (Congdon & Keeton eq. 3.86):

$$V_{\text{eff}}^{\text{photon}}(r) = \frac{L^2}{2r^2} - \frac{GML^2}{c^2r^3}\,. \tag{4.11}$$

Setting $dV_{\text{eff}}/dr = 0$ gives a single circular orbit (Carroll eq. 5.70; Congdon & Keeton eq. 3.84 with ℓ_{crit}):

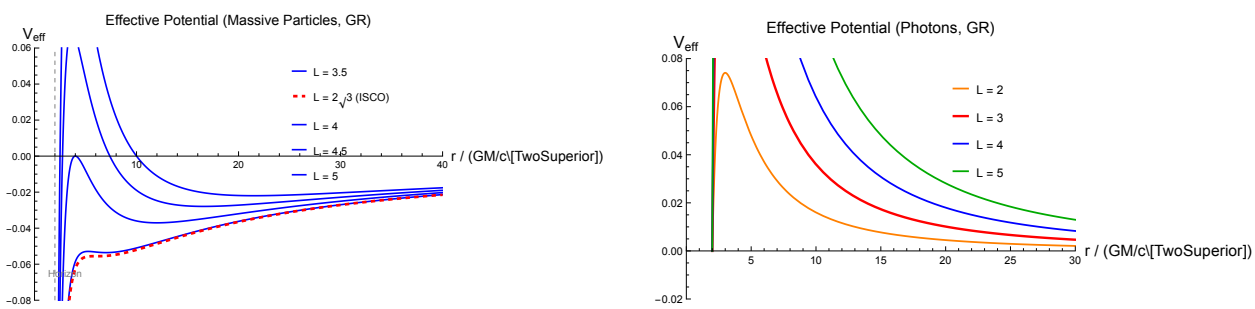
$$\boxed{r_{\text{photon}} = \frac{3GM}{c^2} = \frac{3}{2}\,R_S} \tag{4.12}$$

This is an *unstable* circular orbit — any perturbation causes the photon to either escape to infinity or spiral into the black hole. Nevertheless, the photon sphere plays a key role in strong gravitational lensing near black holes and is directly related to the "photon ring" observed by the Event Horizon Telescope.

4.5 Effective Potential: Newtonian vs. GR

The key qualitative differences between Newtonian and GR effective potentials are:

Feature	**Newtonian**	**GR**
Centrifugal barrier	Diverges as $r \to 0$	Overcome at small r
ISCO	None (stable at any r)	$r = 3R_S$
Photon orbits	None	Unstable at $r = 3R_S/2$
Plunge orbits	Impossible	Possible for $r < r_{\text{ISCO}}$



(a) Massive particles ($\epsilon = 1$) (b) Photons ($\epsilon = 0$)

Figure 4.2: GR effective potentials for several values of angular momentum L (in GM/c units). **Left:** For massive particles, the red dashed curve shows $L = L_{\text{ISCO}} = 2\sqrt{3}\,GM/c$, where the stable and unstable circular orbits merge. **Right:** For photons, the potential always has a maximum at $r = 3GM/c^2$ (the photon sphere). Interactive versions in `effective_potential.nb`.

4.6 Perihelion Precession

In Newtonian gravity, bound orbits are perfect ellipses that close on themselves. In GR, the extra $-GML^2/(c^2r^3)$ term in the effective potential causes the orbit to *precess*: the perihelion (point of closest approach) advances by a small angle each orbit.

Following Carroll Sec. 5.5 and using the substitution $x = L^2/(GMr)$, the orbit equation becomes (Carroll eq. 5.79):

$$\frac{d^2x}{d\phi^2} + x = 1 + \frac{3G^2M^2}{L^2c^2}\,x^2\,. \tag{4.13}$$

The last term is the GR correction. Treating it as a perturbation ($x = x_0 + x_1$, with $x_0 = 1 + e\cos\phi$ the Keplerian solution), one finds the accumulated precession per orbit

(Carroll eq. 5.92; Congdon & Keeton eq. 3.95):

$$\boxed{\Delta\phi = \frac{6\pi GM}{c^2 a(1-e^2)} = \frac{6\pi GM}{L^2/(GM)}} \tag{4.14}$$

where a is the semi-major axis and e the eccentricity.

For Mercury (Carroll eq. 5.96):

$$\Delta\phi_{\text{Mercury}} = 5.01 \times 10^{-7}\ \text{rad/orbit} = 0.103''/\text{orbit} = 43.0''/\text{century}\,. \tag{4.15}$$

This matched the previously unexplained 43 arcseconds/century discrepancy in Mercury's orbit and was one of Einstein's first confirmations of GR.

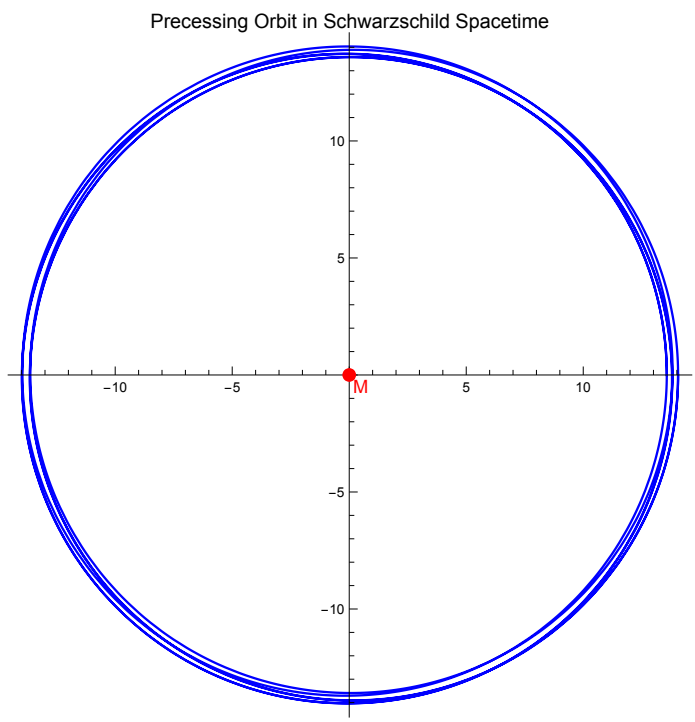


Figure 4.3: A precessing orbit in Schwarzschild spacetime, computed by numerically integrating the orbit equation. The perihelion advances by a small angle each orbit, tracing out a rosette pattern. (cf. Carroll Fig. 5.6)

4.7 Null Geodesics and the Deflection of Light (Preview)

For photons, the orbit equation (using $u = 1/r$) becomes (Congdon & Keeton eq. 3.87):

$$\frac{d\phi}{dr} = \pm\frac{L}{r^2}\left[\frac{E^2}{c^2} - \frac{L^2}{r^2}\left(1 - \frac{R_S}{r}\right)\right]^{-1/2}. \tag{4.16}$$

A photon approaching a mass M with impact parameter $b = cL/E$ (the perpendicular distance from the mass in the absence of deflection) will be deflected by an angle (Congdon

& Keeton eq. 3.96):

$$\hat{\alpha} = \frac{4GM}{c^2 b}\,, \tag{4.17}$$

in the weak-field limit ($b \gg R_S$). This is *twice* the Newtonian prediction and was confirmed by Eddington's 1919 solar eclipse observations.

The full derivation of the deflection angle is the subject of Module 2. We mention it here because the effective potential framework makes it clear *why* light bends: photons follow null geodesics of the curved Schwarzschild spacetime, and the GR correction term in the effective potential deflects them more strongly than Newtonian gravity would predict.

4.8 Brief Overview: The Kerr Metric

Real astrophysical black holes are not perfectly described by the Schwarzschild solution because they *rotate*. The metric for a rotating, axially symmetric black hole is the **Kerr metric**, discovered by Roy Kerr in 1963. In Boyer–Lindquist coordinates, it takes the form:

$$ds^2 = -\left(1 - \frac{R_S r}{\Sigma}\right) c^2 dt^2 - \frac{2R_S r a \sin^2\theta}{\Sigma}\, c\, dt\, d\phi + \frac{\Sigma}{\Delta}\, dr^2 + \Sigma\, d\theta^2 + \frac{A \sin^2\theta}{\Sigma}\, d\phi^2\,, \tag{4.18}$$

where

$$\Sigma = r^2 + a^2\cos^2\theta\,, \qquad \Delta = r^2 - R_S r + a^2\,, \qquad A = (r^2 + a^2)^2 - a^2\Delta\sin^2\theta\,, \tag{4.19}$$

and $a = J/(Mc)$ is the **spin parameter** (angular momentum per unit mass, with dimensions of length). Note $0 \le a \le GM/c^2$ for a physical black hole.

Key features that differ from Schwarzschild:

- **Frame dragging:** The $dt\,d\phi$ cross-term forces particles near the black hole to co-rotate. This effect is absent in Schwarzschild.
- **Ergosphere:** The region $R_S/2 < r < (R_S + \sqrt{R_S^2 - 4a^2\cos^2\theta})/2$ where even stationary observers must rotate with the black hole.
- **Spin-dependent ISCO:** The ISCO radius depends on the spin parameter and the direction of the orbit (prograde vs. retrograde). For a maximally spinning black hole ($a = GM/c^2$): $r_{\rm ISCO} = GM/c^2$ (prograde) vs. $r_{\rm ISCO} = 9GM/c^2$ (retrograde), compared to $6GM/c^2$ for Schwarzschild.
- **Two horizons:** $r_\pm = GM/c^2 \pm \sqrt{(GM/c^2)^2 - a^2}$.

Remark 4.8.1. For gravitational lensing by galaxies and clusters, the Kerr metric is not needed: the lensing effect is dominated by the total enclosed mass, and the weak-field/thin-lens approximation (Module 1e and beyond) washes out spin effects at the angular scales of interest. However, Kerr lensing is an active research topic for *strong* lensing near individual black holes — for instance, the "photon ring" structure observed by the Event Horizon Telescope, where frame-dragging produces asymmetric brightness patterns.

4.9 Summary

- Killing vectors (∂_t, ∂_ϕ) yield conserved energy E and angular momentum L along geodesics.
- The radial motion reduces to a 1D effective potential problem: $V_{\text{eff}} = -\epsilon GM/r + L^2/(2r^2) - GML^2/(c^2r^3)$.
- The GR correction term $\propto 1/r^3$ is responsible for: ISCO at $r = 3R_S$, photon sphere at $r = 3R_S/2$, perihelion precession, and enhanced light deflection.
- Mercury's perihelion precesses by 43 arcsec/century — confirmed by observation and one of the first tests of GR.
- The Kerr metric generalizes Schwarzschild to rotating black holes (frame dragging, ergosphere, spin-dependent ISCO), but is not needed for galaxy/cluster-scale lensing.

4.10 Problems

Exercise 4.10.1. Starting from the normalization condition (4.5) and the conserved quantities E and L, derive the effective potential (4.7) for massive particles ($\epsilon = 1$). Show that in the Newtonian limit ($c \to \infty$, $R_S \to 0$), it reduces to $V_{\text{eff}}^{\text{Newton}} = -GM/r + L^2/(2r^2)$.

Exercise 4.10.2. Derive the ISCO radius by finding the value of L for which the two circular orbit solutions (4.9) coincide (i.e., the discriminant vanishes). Verify $r_{\text{ISCO}} = 6GM/c^2 = 3R_S$.

Exercise 4.10.3. Show that the only circular photon orbit in the Schwarzschild spacetime is at $r = 3GM/c^2 = 3R_S/2$, and that it is unstable (a maximum of $V_{\text{eff}}^{\text{photon}}$).

Exercise 4.10.4. Compute the perihelion precession of Mercury using eq. (4.14) with: $M = M_\odot$, $a = 5.79 \times 10^{10}$ m, $e = 0.2056$. Verify $\Delta\phi \approx 43''$/century (Mercury's orbital period is 87.97 days).

Exercise 4.10.5. Using the effective potential for massive particles with $L = 4GM/c$:

(a) Plot $V_{\text{eff}}(r)$ and identify the stable circular orbit radius, the unstable circular orbit radius, and the ISCO.

(b) For a particle with $\mathcal{E}$ slightly above $V_{\text{eff}}(r_+)$, describe qualitatively the type of orbit (bound precessing ellipse, scatter orbit, or plunge orbit).

4.11 Solutions to Exercises

Full worked solutions are also available in the Mathematica script `problems_01d.wl`, which verifies each answer symbolically.

Exercise 4.10.1: Deriving the Effective Potential

We start from the normalization condition (4.5) with $f(r) = 1 - R_S/r$:

$$-f\left(\frac{dt}{d\lambda}\right)^2 + \frac{1}{f}\left(\frac{dr}{d\lambda}\right)^2 + r^2\left(\frac{d\phi}{d\lambda}\right)^2 = -\epsilon\,.$$

Substituting the conserved quantities $E = f\,dt/d\lambda$ and $L = r^2\,d\phi/d\lambda$:

$$-\frac{E^2}{f} + \frac{1}{f}\left(\frac{dr}{d\lambda}\right)^2 + \frac{L^2}{r^2} = -\epsilon\,.$$

Multiplying through by f and rearranging:

$$\begin{aligned}\left(\frac{dr}{d\lambda}\right)^2 &= E^2 - f(r)\left(\epsilon + \frac{L^2}{r^2}\right)\\ &= E^2 - \left(1 - \frac{R_S}{r}\right)\left(\epsilon + \frac{L^2}{r^2}\right)\\ &= E^2 - \epsilon + \frac{\epsilon R_S}{r} - \frac{L^2}{r^2} + \frac{R_S L^2}{r^3}\,.\end{aligned}$$

Dividing by 2 and identifying terms:

$$\frac{1}{2}\left(\frac{dr}{d\lambda}\right)^2 + \underbrace{\left(-\frac{\epsilon GM}{r} + \frac{L^2}{2r^2} - \frac{GML^2}{c^2r^3}\right)}_{V_{\text{eff}}(r)} = \frac{1}{2}(E^2 - \epsilon) \equiv \mathcal{E}\,,$$

where we used $R_S = 2GM/c^2$. This gives the effective potential (4.7):

$$\boxed{V_{\text{eff}}(r) = -\frac{\epsilon\,GM}{r} + \frac{L^2}{2r^2} - \frac{GML^2}{c^2r^3}}$$

Newtonian limit: The last term $-GML^2/(c^2r^3)$ is the GR correction. In the Newtonian limit ($c \to \infty$, or equivalently $R_S \to 0$), this term vanishes, leaving:

$$V_{\text{eff}}^{\text{Newton}} = -\frac{GM}{r} + \frac{L^2}{2r^2}\,,$$

which is the standard Newtonian effective potential. The centrifugal barrier $L^2/(2r^2)$ diverges as $r \to 0$, preventing any particle from reaching the origin. The GR correction term is attractive at small r and allows the barrier to be overcome — this is why particles can plunge into black holes.

Exercise 4.10.2: ISCO Derivation

Working in $G = c = M = 1$ units, the effective potential for massive particles is $V_{\text{eff}} = -1/r + L^2/(2r^2) - L^2/r^3$.

Step 1: Circular orbit condition. Setting $dV_{\text{eff}}/dr = 0$:

$$\frac{1}{r^2} - \frac{L^2}{r^3} + \frac{3L^2}{r^4} = 0 \quad \Longrightarrow \quad r^2 - L^2 r + 3L^2 = 0\,.$$

The two solutions are:

$$r_\pm = \frac{L^2 \pm \sqrt{L^4 - 12L^2}}{2} = \frac{L^2}{2}\left(1 \pm \sqrt{1 - \frac{12}{L^2}}\right)\,.$$

Step 2: ISCO condition. The ISCO occurs when the two circular orbits merge, i.e., when the discriminant vanishes:

$$L^4 - 12L^2 = 0 \quad \Longrightarrow \quad L^2(L^2 - 12) = 0 \quad \Longrightarrow \quad L = 2\sqrt{3}\,.$$

Step 3: ISCO radius. Substituting $L^2 = 12$:

$$r_{\text{ISCO}} = \frac{12}{2} = \boxed{6} \quad \left(= \frac{6GM}{c^2} = 3\,R_S\right)\,.$$

Verification via $d^2V/dr^2 = 0$: Alternatively, we can solve $dV/dr = 0$ and $d^2V/dr^2 = 0$ simultaneously:

$$\frac{d^2V_{\text{eff}}}{dr^2} = -\frac{2}{r^3} + \frac{3L^2}{r^4} - \frac{12L^2}{r^5} = 0\,.$$

Combined with $dV/dr = 0$, this yields $r = 6$, $L = 2\sqrt{3}$, confirming the result.

Exercise 4.10.3: Photon Sphere

For photons ($\epsilon = 0$), the effective potential is (in $G = c = M = 1$ units):

$$V_{\text{eff}}^{\text{photon}} = \frac{L^2}{2r^2} - \frac{L^2}{r^3}\,.$$

Circular orbit: Setting $dV/dr = 0$:

$$-\frac{L^2}{r^3} + \frac{3L^2}{r^4} = 0 \quad \Longrightarrow \quad \frac{L^2}{r^4}(3 - r) = 0 \quad \Longrightarrow \quad \boxed{r = 3} \quad \left(= \frac{3GM}{c^2} = \frac{3}{2}\,R_S\right)\,.$$

Stability: Evaluating the second derivative at $r = 3$:

$$\left.\frac{d^2V_{\text{eff}}}{dr^2}\right|_{r=3} = \left.\frac{3L^2}{r^4} - \frac{12L^2}{r^5}\right|_{r=3} = \frac{3L^2}{81} - \frac{12L^2}{243} = \frac{L^2}{81}\left(3 - \frac{12}{3}\right) = -\frac{L^2}{81}\,(1) = -\frac{2L^2}{81} < 0\,.$$

Since $d^2V/dr^2 < 0$, the effective potential has a **maximum** at $r = 3$, confirming that the photon orbit is **unstable**. Any perturbation causes the photon to spiral inward (capture) or escape to infinity.

Exercise 4.10.4: Mercury's Perihelion Precession

Using eq. (4.14) with Mercury's orbital parameters:

$$G = 6.674 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}} \qquad M_\odot = 1.989 \times 10^{30}\ \mathrm{kg}$$
$$a = 5.79 \times 10^{10}\ \mathrm{m} \qquad e = 0.2056$$

Step 1: Precession per orbit.

$$\begin{aligned}\Delta\phi &= \frac{6\pi GM}{c^2\, a\,(1-e^2)} \\ &= \frac{6\pi \times 6.674\times10^{-11} \times 1.989\times10^{30}}{(2.998\times10^8)^2 \times 5.79\times10^{10} \times (1-0.2056^2)} \\ &= 5.01 \times 10^{-7}\ \mathrm{rad/orbit}\,.\end{aligned}$$

Step 2: Convert to arcseconds.

$$\Delta\phi = 5.01 \times 10^{-7}\ \mathrm{rad} \times \frac{180 \times 3600}{\pi}\ \frac{\mathrm{arcsec}}{\mathrm{rad}} = 0.1034\ \mathrm{arcsec/orbit}\,.$$

Step 3: Orbits per century.

$$N_{\mathrm{orbits}} = \frac{100 \times 365.25}{87.97} = 415.2\ \mathrm{orbits/century}\,.$$

Step 4: Precession per century.

$$\Delta\phi_{\mathrm{century}} = 0.1034 \times 415.2 = \boxed{43.0\ \mathrm{arcsec/century}}\,.$$

This matches the observed anomalous precession of Mercury, confirming one of the three classical tests of general relativity.

Exercise 4.10.5: Effective Potential for $L = 4$

(a) Working in $G = c = M = 1$ units with $L = 4$, the circular orbit radii are found by solving $dV_{\mathrm{eff}}/dr = 0$:

$$r^2 - 16r + 48 = 0 \quad \Longrightarrow \quad r = \frac{16 \pm \sqrt{256-192}}{2} = \frac{16 \pm 8}{2}\,.$$

Therefore:

- **Stable circular orbit** (minimum of V_{eff}): $r_+ = 12$, $\quad V_{\mathrm{eff}}(12) = -1/24 \approx -0.0417$
- **Unstable circular orbit** (maximum of V_{eff}): $r_- = 4$, $\quad V_{\mathrm{eff}}(4) = -1/8 + 1/2 - 1 = -0.0625 + \ldots$

  More precisely: $V_{\mathrm{eff}}(4) = -1/4 + 16/32 - 16/64 = -0.25 + 0.5 - 0.25 = 0$
- **ISCO** (for reference): $r_{\mathrm{ISCO}} = 6$ occurs at $L = 2\sqrt{3} \approx 3.464$, which is below $L = 4$, so the two circular orbits are well separated.

The plot (see `problems_01d.wl`) shows the GR potential (solid) with the Newtonian potential (dashed) for comparison. The GR correction causes $V_{\text{eff}} \to -\infty$ as $r \to 0$, allowing plunge orbits.

(b) For $\mathcal{E}$ slightly above $V_{\text{eff}}(r_+)$, the particle oscillates between a minimum radius (perihelion) and a maximum radius (aphelion), but the orbit does not close due to the GR correction term. The perihelion advances by a small angle each orbit, producing a **bound precessing ellipse** — the same effect responsible for Mercury's perihelion precession (Exercise 4.10.4).

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